

Informal Regulation and Housing Supply: Evidence from Chicago’s Aldermanic Privilege

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Abstract

Land-use regulation in cities depends not only on the formal rules, but also on the individual politicians who administer them. Many U.S. cities allow substantial informal discretion in the development approval process, often exercised by the local politician representing the district where development is proposed. I study informal regulatory discretion in Chicago, where “aldermanic privilege” gives city council members substantial influence over development in their wards. I construct alderman-level “stringency scores” from permit processing times and use complementary event-study and boundary-discontinuity designs to estimate the effects of these informal regulations on housing supply and prices. Assignment to a more stringent alderman lowers permitting activity by 12%, while new multifamily buildings on the more-stringent side of a ward boundary have about 6% fewer dwelling units per acre. I also find suggestive evidence of higher rents and home sale prices on the more-stringent side of ward boundaries. These results suggest that informal regulation distorts housing supply and may affect citywide housing affordability, even in areas where the formal zoning rules are similar.

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1 Introduction

Housing costs in the United States have risen sharply, and a large literature points to land-use regulation as a central cause. Restrictive zoning and permitting frictions limit housing supply and raise prices, and may distort the allocation of labor across cities (Glaeser, Gyourko and Saks, 2005; Saiz, 2010; Turner, Haughwout and Van Der Klaauw, 2014; Hsieh and Moretti, 2019; Duranton and Puga, 2023; Soltas, 2024; Soltas and Gruber, 2026). However, formal zoning rules are not the only determinant of what gets built. Informal regulations, often exercised through the discretionary power of local politicians, also play a significant role in shaping development patterns. Many major cities in the United States have a fragmented regulatory approval system that includes the discretionary judgment of the local city councilmember or a similar politician.¹ This potential for individual discretion to differentiate otherwise similar regulatory environments induces uncertainty in the development process and lowers housing supply in cities (Mayer and Somerville, 2000a; Hilber and Robert-Nicoud, 2013; Khan, 2021; Mast, 2024). Informal discretion from individual politicians can substantially lengthen the development timeline. When developers face long and unpredictable timelines, projects shrink and marginal investments become unviable, which can lead to a reduction in housing supply (Cunningham, 2006; Bloom, 2009; Oh, Yang and Yoon, 2024). Despite this evidence, there is little empirical work on how individual politicians use these informal regulatory powers to shape housing supply.

This paper studies this form of informal regulatory power in the context of Chicago’s “aldermanic privilege,” the unwritten but influential authority of each of Chicago’s 50 city council members (aldermen) to unilaterally block or approve development in that member’s ward. Chicago has no city charter, so there is no statutory basis for this power (McClintock, 2025). Other aldermen vote with the local member on land use matters in exchange for similar support in their own wards, city agencies follow the local alderman’s lead, and devel-

¹Analogous forms of council-member deference in land-use decisions have been documented in Philadelphia, Los Angeles, and New York City (Burnett and Kogan, 2014; Jacobson, 2023; Mitre-Becerril and MacDonald, 2024; New York City Charter Revision Commission, 2025).

opers understand that major projects require the alderman’s blessing ([Thale, 2005](#); [Einstein, Glick and Palmer, 2019](#); [Khan, 2021](#)). This creates sharp changes in regulatory authority at ward boundaries, even when the neighborhood and zoning are quite similar. Two parcels on opposite sides of the same street, in the same neighborhood, with the same zoning, can face very different regulatory regimes depending on which alderman represents them. I use this sudden change in regulatory authority to study how informal regulations affect housing supply and prices.

To measure the regulatory environment each alderman creates, I construct “regulatory stringency scores” from processing times for permits likely to involve substantial aldermanic discretion, which include new construction, renovations, and demolitions. I deem these “high-discretion” permits since they are the most likely to require aldermanic approval. I use processing times as a proxy for regulatory stringency because delay makes projects less valuable. A developer who expects years of meetings, redesigns, and uncertainty may build a smaller project or be deterred from building at all ([Kydland and Prescott, 1982](#); [Mayer and Somerville, 2000a](#); [Paciorek, 2013](#); [Gyourko, Hartley and Krimmel, 2021](#); [Soltas and Gruber, 2026](#)).

I first use the 2015 ward redistricting to test whether permitting changes when a census block is reassigned to a different ward. This design holds location fixed while changing the assigned alderman. Using this within-census-block variation, I find that reassignment toward greater stringency reduces the annual number of high-discretion permits recorded by the city by about 12% during 2015–2020. This effect is asymmetric across stringent and lenient aldermen. Census blocks assigned to a more stringent alderman see a 20.7% reduction in high-discretion permits, compared with a 7.0% increase for blocks reassigned to a more lenient alderman. I find no statistically significant effect on low-discretion permits, consistent with an effect concentrated in permits more exposed to aldermanic discretion.

I then examine whether new housing is built at lower density under more stringent aldermen. Following [Bayer, Ferreira and McMillan \(2007\)](#) and [Kulka, Sood and Chiumenti](#)

(2026), I compare new buildings within 500 feet of the same stretch of a ward boundary, built while the same two aldermen represented the two wards. New multifamily buildings on the more-stringent side have about 6% fewer dwelling units per acre, and buildings with five or more units about 10% fewer, while density across all new construction does not differ.

I also compare rents and sale prices across ward boundaries. Listed rents are about 1.7% higher on the more-stringent side of the boundary. Home sale prices are about 1.1% higher, but the estimate is imprecise and not statistically significant. The permit, density, and price results are similar when I use the size of the stringency difference rather than only its direction. Although these results are more suggestive, their direction is consistent with lower supply on the more-stringent side and with informal regulation affecting affordability.

This paper contributes most directly to work on local political control and fragmented land-use authority. [Fischel \(2009\)](#) argues that homeowners use local institutions to restrict development and protect property values, and subsequent work shows how decentralized jurisdictions can adopt restrictions that are locally rational but collectively costly ([Hilber and Robert-Nicoud, 2013](#); [Ortalo-Magné and Prat, 2014](#); [Monarrez and Schönholzer, 2023](#); [Mast, 2024](#); [Bordeu, 2025](#); [Kulka, Sood and Chiumenti, 2026](#)). I show that politicians within a fragmented system can create different regulatory environments even when formal rules are shared.

In Chicago specifically, [Freemark \(2020\)](#) studies formal upzoning and [Khan \(2021\)](#) finds that aldermen approve smaller rezonings near ward boundaries where the benefits to their constituents are lower. I add to this literature by showing that the effects of informal regulatory discretion extend beyond formal zoning or statutory upzonings and appear in permitting activity and the density of new construction, not only in zoned density.

The paper also contributes to work on permitting delays and time-to-build, since the effects of these informal regulations are often realized through development uncertainty and delays. Development lags are a first-order determinant of housing supply ([Mayer and Somerville, 2000a,b](#); [Paciorek, 2013](#)). Recent work finds that rules-based planning shortens

construction timelines and that developers place substantial value on lots with ready-to-issue permits (Lyons and Sweeney, 2025; Soltas and Gruber, 2026). I use alderman-specific variation in permitting time and connect this to both permit activity and completed housing density.

2 Institutional Background and Data

2.1 Aldermanic Privilege and the Development Process

Chicago’s City Council consists of 50 aldermen, each representing a single ward. Informal norms give each Chicago alderman broad authority over development in the ward. This practice, known as “aldermanic privilege” or “aldermanic prerogative,” rests on custom rather than formal law, in part because Chicago has no city charter (McClintock, 2025). On matters affecting a single ward, other aldermen defer to the local representative, and city agencies follow the alderman’s lead (Thale, 2005; Zhang, 2011). The convention is strong enough that aldermen who also led their ward’s party organization were described as “little mayors” of their wards (Thale, 2005). One alderman described the system candidly: “I often liken the City of Chicago to a feudal system, where the mayor is sort of a de facto king. And each alderman is the lord of their individual fiefdom.” (Keefe, 2013)

The formal mechanism. Two reforms narrowed this authority in 2019. A new construction code repealed a rule that had generally required the Department of Buildings to notify the local alderman of permit applications and wait ten days. Mayor Lightfoot’s Executive Order 2019-2 separately directed city departments not to defer to aldermen when municipal law did not require it. Despite these reforms, aldermen managed to keep much of their influence. Other notification requirements remained, departments still asked for aldermanic input, and zoning still required City Council approval (City of Chicago, 2019; Mayor’s Press Office, 2019; Schutz, 2019). Lightfoot described the problem in 2019: “If you’ve got 50

separate fiefdoms that you have to deal with on issues like affordable housing, or just getting a sign on your building, it makes doing business with the city of Chicago damn near impossible.” (Spielman, 2019b) The Metropolitan Planning Council made the same point from the developer side: “before a development even gets to the point of being indefinitely delayed, many never make it past ‘Go’ because of required aldermanic letters of support.” (Metropolitan Planning Council, 2019)

How prerogative shapes development. Aldermanic discretion enters well before formal permit issuance and continues after a project is filed.

Pre-application. Experienced Chicago developers know that projects requiring rezoning or smaller modifications need the alderman’s support before formal papers are filed. Aldermanic support is not legally required, but developers who proceed without it risk paying for design, legal, and engineering work on a project the alderman later blocks. One anonymous Chicago architect described the dynamic: “To get the alderman, you have to get the neighborhood group, and that’s where the trouble comes in. It’s fraught with risk, even to have this conversation.” (Giangreco, 2024)

Rezoning. For projects requiring a change to the zoning code, the alderman’s role is effectively dispositive. Although zoning amendments formally require a City Council vote, the council “routinely shows unquestioning deference to local aldermen even in the absence of any articulated local concern” (Chase, 2023). Equally consequential is the alderman’s ability to preemptively *downzone* parcels to foreclose future development. Joseph Schwieterman, a DePaul University professor who studies Chicago land use, noted that the practice “creates real uncertainty for real estate developers when downzoning becomes ad hoc or driven by the desire of an alderman to settle scores.” (Gallun, 2019) Documented episodes include Ald. Brendan Reilly’s proposals to downzone a River North site where a developer had proposed a 60-story tower and a separate site to block a 615-room hotel, and a Pilsen downzoning orchestrated by Ald. Solis that stripped a parcel of its residential designation to kill a planned

465-unit apartment project ([Gallun, 2019](#)).

Permit processing. For projects that proceed by right under existing zoning, the alderman retains substantial informal influence over the Department of Buildings permit queue. The well-publicized corruption case against Alderman Ed Burke illustrated the mechanism in an extreme form, where federal prosecutors described the alderman’s staff holding up routine permits, including a driveway permit for a fast-food restaurant, until the owner hired Burke’s law firm ([Spielman, 2019a](#)). The case highlighted how permit timing is a lever aldermen can use to delay and suppress projects they oppose, which is another unwritten rule in Chicago that developers have long understood. In an interview with the author, a former alderman described substantial use of the permit process to deter developers from trying to overrule the alderman on rezoning and planning matters earlier in the process.

A recent example. Developer Fern Hill began community outreach in 2021, proposing a roughly 500-unit mixed-income tower at a transit-accessible site in the 2nd Ward, with approximately 100 affordable units and an estimated \$2.5 million in annual property tax revenue. After four years and more than ten community meetings, Ald. Brian Hopkins announced in January 2025 that he would not advance the project, citing opposition from residents of neighboring high-rises. He was explicit about the source of his authority: “Ultimately I am the decision maker on this. I’m not going to hide from that. This is a zoning decision and in Chicago we have something called aldermanic prerogative.” ([Achong, 2025](#)) Within two weeks the developer resubmitted a 349-unit proposal, cutting 151 units (30 percent) and 101 feet from the tower; Hopkins then approved it ([TRD Staff, 2026](#)). The episode captures both margins this paper studies. The approved project is smaller than what the developer proposed (intensive margin), and the four-year ordeal can signal to other developers the cost of attempting large projects in wards with active aldermanic gatekeeping (extensive margin).

Federal scrutiny. In 2023, the U.S. Department of Housing and Urban Development (HUD) concluded that aldermanic prerogative operates as a “local veto” that allows wards to block, deter, or downsize affordable housing, especially in majority-white areas (Chase, 2023); outside groups have made similar civil-rights arguments (Shriver Center on Poverty Law, 2019). The question of whether to reform or abolish aldermanic privilege remains a central issue in Chicago politics.²

2.2 Building Permits and the Permit Process

The City’s public data contain permits recorded by the Department of Buildings since 2006. Each record reports when the application began and when the permit was issued, but cannot capture projects abandoned before they progress far enough to generate a permit record. I use the universe of “high-discretion” permits from 2006–2022 that have valid geographic coordinates and processing times of at least one day. Online Appendix A.3 describes the separate sample used to estimate stringency scores in the event study. I divide permits into two groups based on how much aldermanic discretion they are likely to face.

High-discretion permits are permits for projects where aldermen are most likely to matter: new construction, renovation or alteration, demolition, porch construction, and reinstated permits.³ These are the permit types most likely to draw community opposition and face delays due to community meetings or direct aldermanic intervention. *Low-discretion permits* are all remaining permit types outside this high-discretion group, excluding signs. I use them as a placebo comparison group because they are generally less exposed to discretionary aldermanic intervention.

High-discretion permits take longer to process and vary more across aldermen. Their mean processing time is 57.3 days, compared with 26.2 days for low-discretion permits (Figure 1). Processing time is negatively associated with ward-month permit volume for high-

²See <https://news.wttw.com/elections/voters-guide/2023/races/city-council>, where each city council candidate was asked specifically about their stance on aldermanic prerogative.

³Porch construction permits have received extra scrutiny since the 2003 balcony collapse.

discretion permits but weakly positively associated for low-discretion permits, consistent with the former capturing regulatory friction rather than general administrative capacity. [Online Appendix Table C.3](#) reports the corresponding statistics.

2.3 Regulatory Stringency Scores

I construct the stringency score using permit records from 2006–2022. I first adjust processing times for observed differences across wards, locations, permits, and months, then estimate the remaining alderman-specific differences. The first stage regresses log processing time at the permit level on ward-level demographic controls (median household income, racial composition, homeownership rate, population), permit-location geographic controls (distance to the central business district (CBD), lakefront proximity, Chicago Transit Authority (CTA) rail access), lagged ward-month permit volume, and fixed effects for year-month, permit type, and review type:

$$\log(\text{Processing Time}_i) = X_i'\theta + \phi_{m(i)} + \psi_{c(i)} + \xi_{v(i)} + \varepsilon_i, \quad (1)$$

where i indexes permits, X_i contains the observed permit and location controls, and $m(i)$, $c(i)$, and $v(i)$ denote the permit’s year-month, permit class, and review type. [Online Appendix Table B.1](#) reports the coefficients on the first-stage controls.

The second stage averages the adjusted processing times by ward and month, then regresses those averages on alderman indicators and the same lagged ward-month permit-volume control. Ward-months are weighted by the number of permits observed:

$$\bar{\varepsilon}_{wm} = \mu + \alpha_{a(w,m)} + \chi \text{Volume}_{w,m-1} + \nu_{wm}. \quad (2)$$

Here $a(w, m)$ denotes the alderman serving ward w in month m . The raw alderman fixed effects $\hat{\alpha}_a$ can be noisy for aldermen with few permits. I therefore apply empirical Bayes shrinkage ([Kline, Rose and Walters, 2022, 2024](#)) to pull imprecise estimates toward the

grand mean, then standardize the scores, denoted S_a , to have mean zero and unit standard deviation.

[Online Appendix Figure B.1](#) maps the stringency scores in 2014 and 2022, shows their distribution, and reports a low within-ward correlation between predecessor and successor aldermen ($\rho \approx 0.16$). The scores vary across wealthy and poor neighborhoods and across the North and South Sides, supporting an alderman-specific rather than ward-level interpretation.

I validate the score using aldermen’s public statements on aldermanic prerogative from the 2019 *Chicago Sun-Times* candidate questionnaire, the 2019 WBEZ mayoral questionnaire, the 2023 WTTW voters’ guides, and news coverage.⁴ Aldermen who publicly supported reforming or abolishing prerogative have lower scores, and those who supported keeping it have higher scores ([Online Appendix Table B.2](#)). There appears to be a moderate correlation between differences in my stringency metric and views on prerogative.

2.4 New Construction Data

I construct a citywide dataset of new residential buildings in Chicago permitted between 2006 and 2022. I start from every City of Chicago building permit for a new residential building and match each permit to the Cook County Assessor’s records of the building it produced: residential improvement records, condominium records, and, from 2021, commercial apartment valuations. These records report dwelling units, building area, and land area. I date each building by its first permit’s issue date. Each building is assigned to the ward and alderman in office on that date. For the boundary-discontinuity analyses, I divide ward boundaries into approximately 1,320-foot segments and assign each building to the nearest

⁴The *Sun-Times* published each candidate’s questionnaire on a separate page (for example, [Walter Burnett Jr., 27th Ward](#)). The other sources are the [WBEZ 2019 mayoral questionnaire](#), the [WTTW 2023 voters’ guide](#), and news coverage in WTTW ([March 2019](#), [April 2019](#), [May 2019](#)), the *Sun-Times* ([May 2019](#), [January 2020](#), [February 2020](#), [December 2020](#), [July 2021](#), [March 2022](#)), the *Chicago Reader* ([January 2018](#), [November 2022](#)), *Block Club Chicago* ([November 2019](#)), *The Real Deal* ([December 2019](#)), *Streetsblog Chicago* ([September 2021](#)), *Nadig Newspapers* ([December 2021](#)), PBS ([April 2026](#)), and the [Mayor’s Office](#) ([May 2019](#)).

segment. Because some ward boundaries are several miles long, the segments ensure that compared buildings are near both the same boundary and each other. [Online Appendix A.4](#) describes the matching rules and measurement checks.

Because zoning measured today may reflect subsequent changes, I assign each building the broad zoning group in force on its construction date, using the historical zoning reconstruction described in the same appendix.

I measure density as Dwelling Units Per Acre (DUPAC), defined as dwelling units divided by lot acreage. I measure each building’s distance to its nearest ward boundary.

Table 1 presents summary statistics. The descriptive sample contains 15,350 new residential buildings throughout Chicago, including 5,322 multifamily buildings and 1,977 buildings with five or more units. I report the multifamily samples separately because aldermanic discretion is likely to matter most for larger projects.

2.5 Rental Listings

RentHub provides scraped rental listings. I use Chicago listings from 2014 through December 2022 and collapse repeated scrapes to one floorplan-month observation, using the median asking rent observed that month. Rents are expressed in 2022 dollars using the monthly Chicago CPI-U.

2.6 Home Sales

Home sales come from Cook County Assessor records. I use filtered transactions of single-family homes, townhouses, and small multifamily buildings (2–6 units) from 2006 through 2022. Prices are expressed in 2022 dollars using the monthly Chicago CPI-U. Full 14-digit Property Index Numbers (PINs) determine location and date-specific ward assignments, and same-year Assessor records provide property characteristics. I exclude transactions whose exact day is unavailable when an adjoining ward changes aldermen or the ward map changes during the reported month.

[Online Appendix A](#) provides additional detail on the rental and home sales data.

3 Conceptual Framework

3.1 Developer Problem

I formalize the cost of delay with a simple time-to-build problem ([Kydland and Prescott, 1982](#)).

Consider a developer choosing the scale $q \geq 0$ (units of housing) of a project on a given site. Construction costs are $C(q) = F\mathbf{1}\{q > 0\} + cq + \frac{\eta}{2}q^2$ with $F, \eta > 0$, so building requires a fixed cost and marginal costs rise with project size. The developer can instead leave the site unbuilt and earn zero. Let P denote the local price level, taken as given at the project level.

Let S index the stringency of the local regulatory environment. Stringency maps into an *effective time-to-build* $T(S)$ with $T'(S) > 0$. The term $T(S)$ summarizes entitlement delay, redesign requirements, negotiation costs, financing frictions, and regulatory uncertainty that push expected revenue further into the future. If revenue arrives upon project completion, the developer's expected discounted revenue per unit is $e^{-rT(S)}P$, where r is the relevant discount or financing rate. The developer solves

$$\max_{q \geq 0} e^{-rT(S)}Pq - C(q).$$

Conditional on building, the first-order condition gives the interior scale

$$q^I(S, P) = \frac{e^{-rT(S)}P - c}{\eta}.$$

At this scale, the project's profit is $(e^{-rT(S)}P - c)^2 / (2\eta) - F$. The developer therefore builds

only when the project covers its fixed cost, and the optimal choice is

$$q^*(S, P) = \begin{cases} \frac{e^{-rT(S)}P - c}{\eta}, & \text{if } e^{-rT(S)}P - c \geq \sqrt{2\eta F}, \\ 0, & \text{otherwise.} \end{cases}$$

For projects that are built, $\partial q^*/\partial T(S) < 0$. Longer or more burdensome regulatory processes therefore reduce project scale and can also push marginal projects from “build” to “do not build.”

3.2 From Developer Choices to Empirical Predictions

Indexing sites by i , total housing supply is $Q = \sum_i q_i^*(S_i, P_i)$. Higher $T(S_i)$ reduces Q through two direct margins. Projects that are built are built at lower density, and marginal projects are not built at all. These are the main predictions I take to the data. The flow of permits requiring aldermanic review should fall when stringency increases (Section 4.1), and sites in more stringent regulatory environments should have lower completed density (Section 4.2).

With downward-sloping demand $P'(Q) < 0$, a supply contraction can also raise housing costs. This prediction is harder to test because prices may adjust over a wider area, but Section 4.3 finds suggestive evidence that prices are higher on the more-stringent side of ward boundaries.

3.3 Interpretation of the Stringency Score

In the model, these frictions enter through the index $rT(S)$. A higher value means revenue arrives later and is worth less today. That affects both the developer’s intensive-margin density choice and the extensive-margin decision to build or not.

The aldermanic stringency score S_a constructed in Section 2.3 measures the alderman-specific component of observed processing time. I interpret a higher S_a as a more stringent

regulatory environment and thus a higher effective time-to-build $T(S)$. Other dimensions of regulatory friction, such as impact fees, design mandates, or community benefit requirements, would enter as additional costs in $C(\cdot)$ and reinforce the same qualitative predictions. I treat longer processing times for larger discretionary projects as a noisy signal of the delay that lowers density and aggregate supply.

4 Results

4.1 Permit Activity After Redistricting

I first ask whether permit activity changes when the same location is assigned to a different alderman. Chicago’s 2015 ward redistricting moved some census blocks toward wards represented by more stringent aldermen and others toward wards represented by more lenient aldermen. Nearby blocks that remained in their original ward provide the comparison group. The design therefore holds the physical location fixed while changing its assigned regulatory environment. The primary analysis sample keeps boundaries for which both the origin- and destination-ward aldermen serving in 2014 remained in office when the new map took effect, since newly elected aldermen’s development preferences could be endogenous to prior development activity in the ward. I also require each block to have at least one high-discretion permit during the five years before redistricting. Without that restriction, any later permit on a block with no previous activity would mechanically appear as an increase.

The outcome is the number of high-discretion permits recorded on each census block in each year from 2010 through 2020. The outcome therefore measures when projects observed in the permit data first appeared in the process. These are the high-discretion permits defined in Section 2.2.

I estimate a Poisson pseudo-maximum likelihood event study that treats moves toward

greater stringency and greater leniency as opposite changes:

$$\log \mathbb{E}[Y_{by} \mid \cdot] = \mu_b + \lambda_{g(b),y} + \sum_{t \neq -1} \beta_t Z_b \mathbf{1}\{y - 2015 = t\}. \quad (3)$$

Here Y_{by} is the permit count for block b in calendar year y , μ_b is a block fixed effect, and $\lambda_{g(b),y}$ is a ward-pair-by-year fixed effect for the ward pair $g(b)$ containing block b . Event time is $t = y - 2015$, with $t = -1$ omitted. Let ΔS_b be the destination-minus-origin difference in standardized aldermanic stringency for block b . For blocks that change wards, $Z_b = \text{sign}(\Delta S_b)$; it equals zero for blocks that remain in the same ward. I classify moves using the 2014 incumbents in each block’s origin and destination wards, with scores estimated only from permits filed between 2006 and 2014. Using the 2014 incumbents separates ward reassignment from changes in representation associated with the 2015 elections. This specification imposes equal-and-opposite effects in logs for assignments toward more stringent and more lenient aldermen. Panel A of Figure 2 plots the coefficients β_t . The pre-trends are not distinguishable from zero. The pooled 2015–2020 estimate is -0.127 ($\text{SE} = 0.049$), equivalent to an 11.9% reduction in the annual number of high-discretion permits associated with a move toward greater stringency. Panels B and C relax the symmetry restriction by estimating the two directions separately. The decline in permitting activity comes mostly from moves toward more stringent aldermen, which reduce high-discretion permits by 20.7%; blocks assigned toward more lenient aldermen experience a 7.0% increase. [Online Appendix Table C.2](#) reports the corresponding pooled estimates without the symmetry restriction and shows the results are similar.

This asymmetry suggests that the effect comes from stringent aldermen blocking projects that would otherwise go forward, and less from lenient aldermen approving marginal projects that otherwise would not have gone forward. This distinction matters for the likely aggregate effect of informal regulatory discretion on housing supply because it is possible ex ante that a few particularly pro-development aldermen approve more housing than a centralized system

would. The results instead suggest that informal aldermanic discretion reduces housing supply relative to a more centralized counterfactual.

Many of these blocks' wards elected new aldermen in 2019, so the last years of the sample partly reflect aldermen other than the 2014 incumbents used to classify each move. Restricting the sample to a symmetric 2012–2018 window before those elections ([Online Appendix Figure C.2](#)), or adding low-discretion permits on the same blocks as a comparison group ([Online Appendix Figure C.4](#)), gives similar but noisier estimates, and the decline remains concentrated in moves toward more stringent aldermen. Using the size of the score change rather than its sign gives similar results ([Online Appendix Figure C.3](#)).

4.2 Density at Ward Boundaries

I next examine the new buildings described in [Section 2.4](#) on opposite sides of the same ward boundary, moving the unit of analysis from census blocks to buildings. Buildings within 500 ft of the same boundary segment face similar neighborhood amenities and demand conditions, but different aldermen. Following [Bayer, Ferreira and McMillan \(2007\)](#) and [Kulka, Sood and Chiumenti \(2026\)](#), I divide signed distance to the boundary into 100 ft bands and compare buildings in each band with buildings in the nearest band on the less-stringent side. I estimate the following specification:

$$\log(\text{DUPAC}_{isj}) = \sum_{k \in \mathcal{K}, k \neq [-100, 0)} \kappa_k \mathbf{1}\{d_{is} \in k\} + X_i' \theta + \zeta_{z(i)} + \delta_{sj} + \varepsilon_{isj}. \quad (4)$$

Here DUPAC_{isj} is dwelling units per acre for building i , assigned to boundary segment s and built during joint-service period j , an uninterrupted interval during which the same two aldermen represent the two wards of the boundary. The running variable d_{is} is negative on the less-stringent side and positive on the more-stringent side. The set $\mathcal{K}$ contains ten 100 ft bands from -500 to 500 ft. The band from -100 to 0 ft is omitted, so $\kappa_{[0, 100)}$ compares buildings immediately across the boundary. The vector X_i contains the White

population share, the Black population share, median household income, the bachelor’s-degree share, and the homeownership rate on the building’s side of the boundary. The model also includes fixed effects $\zeta_{z(i)}$ for the zoning group in force when the building was built, and boundary-segment-by-joint-service fixed effects δ_{sj} , which compare buildings near the same short stretch of boundary built while the same two aldermen served. My main estimate is the average difference between the two sides, which replaces the distance bands with a single indicator for the more-stringent side of the 500 ft window. Standard errors are clustered by ward pair.

Figure 3 plots the distance-band estimates. The omitted band is normalized to zero, and the subtitle in each panel reports the average difference. Across all new construction, density does not differ between the two sides (0.002, SE = 0.029). New multifamily buildings on the more-stringent side have 5.9% fewer dwelling units per acre (−0.061, SE = 0.021), and buildings with five or more units have 9.5% fewer (−0.100, SE = 0.059). The difference between the two bands immediately across the boundary is larger ([Online Appendix Table D.1](#)), and weighting each boundary by the size of the stringency difference between its two aldermen gives similar results ([Online Appendix Figure D.3](#)).

As a falsification test, I repeat the same comparison at artificial cutoffs 1,000 ft inside either side of the true ward boundary. These locations do not mark a change in aldermanic authority. [Online Appendix Figure D.1](#) shows little evidence of corresponding breaks at these artificial cutoffs. None of the placebo estimates is statistically distinguishable from zero. [Online Appendix D](#) presents the placebo figures along with continuity tests for fixed location and neighborhood characteristics, restrictions based on boundary geometry, and donut checks.

4.3 Evidence on Rents and Sale Prices

The permit and density results indicate that more stringent aldermen reduce development activity and the density of completed housing. The model in [Section 3](#) indicates that this

should feed into housing costs, the most directly policy-relevant outcome given Chicago’s current rent and price growth. For rents and sales, I estimate

$$\log(Y_{isp}) = \sum_{k \in \mathcal{K}, k \neq [-100, 0)} \kappa_k \mathbf{1}\{d_{is} \in k\} + X_i' \theta + \lambda_{t(i)} + \delta_{sp} + \varepsilon_{isp}, \quad (5)$$

where Y_{isp} is the real listed rent or real sale price for observation i near boundary segment s in calendar period p . As in the density analysis, $\mathcal{K}$ contains ten 100 ft distance bands and the band from -100 to 0 ft is omitted. The coefficient $\kappa_{[0, 100)}$ therefore compares observations in the two bands immediately across the boundary. The main estimate is again the average difference, which replaces the distance bands with an indicator for the more-stringent side of the 500 ft window. The fixed effects compare observations near the same boundary segment in the same month for rents and the same quarter for sales. X_i includes unit or property characteristics and distances to nearby schools, parks, major roads, CTA stops, and Lake Michigan. The term $\lambda_{t(i)}$ denotes property-type fixed effects, which are the listing’s building type for rents and the Assessor property class for sales.

Standard errors are clustered by ward pair.

Figure 4 plots these distance-band estimates, and the subtitles report the average difference.

Listed rents are 1.7% higher on the more-stringent side relative to the less-stringent side (0.017, SE = 0.010), a difference statistically distinguishable from zero at the 10% level. Home sale prices are 1.1% higher (0.011, SE = 0.007), and the estimate is not statistically distinguishable from zero. Weighting each boundary by the size of the stringency difference gives similar results (Online Appendix Figure E.3). Online Appendix E shows that the placebo estimates at artificial cutoffs inside either ward are statistically insignificant and that the result is similar when I restrict the sample to straight boundaries or exclude observations immediately next to the boundary.

5 Discussion

5.1 Interpretation and Mechanisms

Taken together, the redistricting and boundary results show that individual politicians create different regulatory environments within a single city. Within-census-block variation shows that permitting falls when a census block is reassigned to a more stringent alderman, while new multifamily buildings on the more-stringent side of ward boundaries are built at lower density. The price and rent comparisons are more tentative, but their direction is consistent with these differences in housing supply.

The central implication of these results is that the effect of political fragmentation on housing supply extends beyond formal zoning rules. Two otherwise similar projects governed by the same municipal code can face different timelines and prospects for approval simply because they are represented by different aldermen. Political fragmentation can therefore arise within a city, even without separate municipal governments or formally different regulatory codes.

These informal regulations can affect development along both extensive and intensive margins, since additional community meetings and uncertain approval timelines reduce the expected return to development. Developers may respond to these additional demands, or the threat of them, by shrinking or abandoning projects before they enter the permitting data. The Fern Hill episode described in Section 2 illustrates these margins. The project that eventually received approval was smaller than the original proposal, and approval followed four years of meetings and negotiation. In the time-to-build framework of Section 3, delay lowers the value of development, so projects that remain viable are smaller and marginal projects are not built.

These findings complement [Khan \(2021\)](#), who shows that Chicago aldermen approve smaller increases in allowed density near ward boundaries, where their constituents capture fewer of the benefits from development. Informal political control can also shape whether

development proceeds and how much gets built on a site even when the zoning code and neighborhood characteristics are similar.

5.2 Informal Discretion Beyond Chicago

Chicago provides an especially clear setting because aldermanic privilege is unusually powerful, but the underlying institution is not unique to Chicago. Local council-member deference in land-use decisions has also been documented in Philadelphia, Los Angeles, and New York City ([The Pew Charitable Trusts, 2015](#); [Ryu, 2020](#); [Anbinder, 2025](#); [New York City Charter Revision Commission, 2025](#); [Mitre-Becerril and MacDonald, 2024](#); [Burnett and Kogan, 2014](#); [Jacobson, 2023](#)).

In many major U.S. cities, individual politicians can create distinct regulatory environments when projects require discretionary approval and other officials defer to the representative of the affected district. Under these conditions, the identity and preferences of the local politician can matter even when the written rules remain similar across districts. The results here suggest that informal regulatory discretion can dampen housing supply and worsen affordability in other cities as well.

5.3 Policy Implications

Local politicians may possess useful information about neighborhood conditions and provide residents with a clear channel for political accountability. But ward-level representatives also have little reason to internalize the citywide benefits of additional housing, and informal discretion allows them to impose delay and uncertainty after the formal rules have been established. The resulting regulatory environment may reflect the preferences of an individual officeholder or a small group of constituents, without accounting for benefits elsewhere in the city.

While this paper cannot pin down the optimal balance between local input and centralized authority, it does point to predictability and transparency as important goals for reform.

Expanding the scope of by-right approval would allow projects that comply with established rules to proceed without obtaining the support of an individual alderman. Published approval criteria, binding timelines on aldermanic decisions, and meaningful appeal procedures could preserve local participation while limiting uncertainty. Moving some decisions to city-wide bodies could also make it easier to account for housing benefits that extend beyond the ward in which a project is located.

Chicago’s lack of a city charter makes the allocation of authority particularly dependent on unwritten norms. A charter would not by itself eliminate aldermanic influence, but it could formally define approval authority, procedures, and avenues for appeal (McClintock, 2025). More generally, reforms aimed only at zoning classifications or formal voting rules may have limited effects if individual politicians retain informal control over permitting, project design, and administrative implementation.

These institutional choices matter as Chicago confronts rapidly rising rents (Davis, 2026). The results in this paper show that evaluations of land-use and permitting reform proposals should consider how the rules are administered along with the changes being proposed to the statutory rules. When individual political actors possess substantial discretionary authority, who administers the rules can matter as much as the rules themselves.

6 Conclusion

This paper studies how informal regulations and the discretion of individual political actors affect housing supply within a city. I focus on Chicago’s aldermanic privilege, the informal power of each of the city’s 50 aldermen to block or approve development in that member’s ward. This institution allows individual politicians to create different regulatory environments within a single municipal system.

Using permit processing times for projects where aldermen are most likely to matter, I construct aldermanic stringency scores that isolate the alderman-specific component of

regulatory delay after accounting for permit, location, time, and ward characteristics. Using the 2015 ward redistricting to identify changes in assigned stringency, I find that reassignment toward greater stringency reduces the annual number of high-discretion permits recorded by the city by about 12% during 2015–2020. A boundary comparison shows that new multifamily buildings on the more-stringent side of ward boundaries have about 6% fewer dwelling units per acre. I also find listed rents about 1.7% higher on the more-stringent side and a smaller, imprecise 1.1% difference in home prices.

Formal zoning rules leave out part of the regulatory environment a developer faces. Similar projects governed by the same municipal and zoning codes can face different timelines and prospects for approval depending on the local politician. Effective land-use reforms must address not only the formal rules governing what can be built, but also the informal political institutions that determine whether and how development proceeds.

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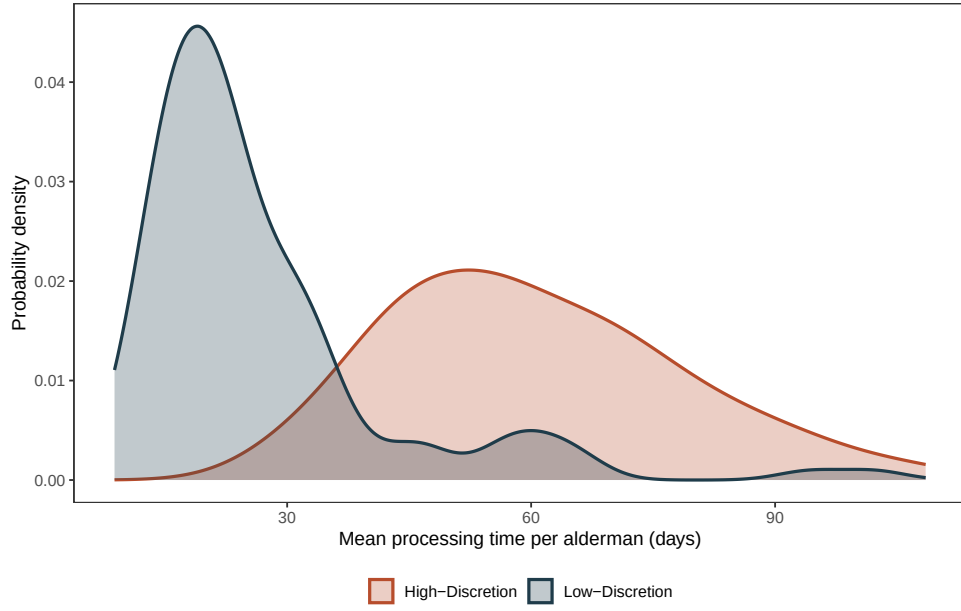


Figure 1: Permit Processing Times Across Aldermen

Notes: The curves show how mean processing times vary across aldermen, separately for high- and low-discretion permits, using records through December 2022. The high- and low-discretion permit groups are defined in Section 2.2. Processing time is days from application to issuance.

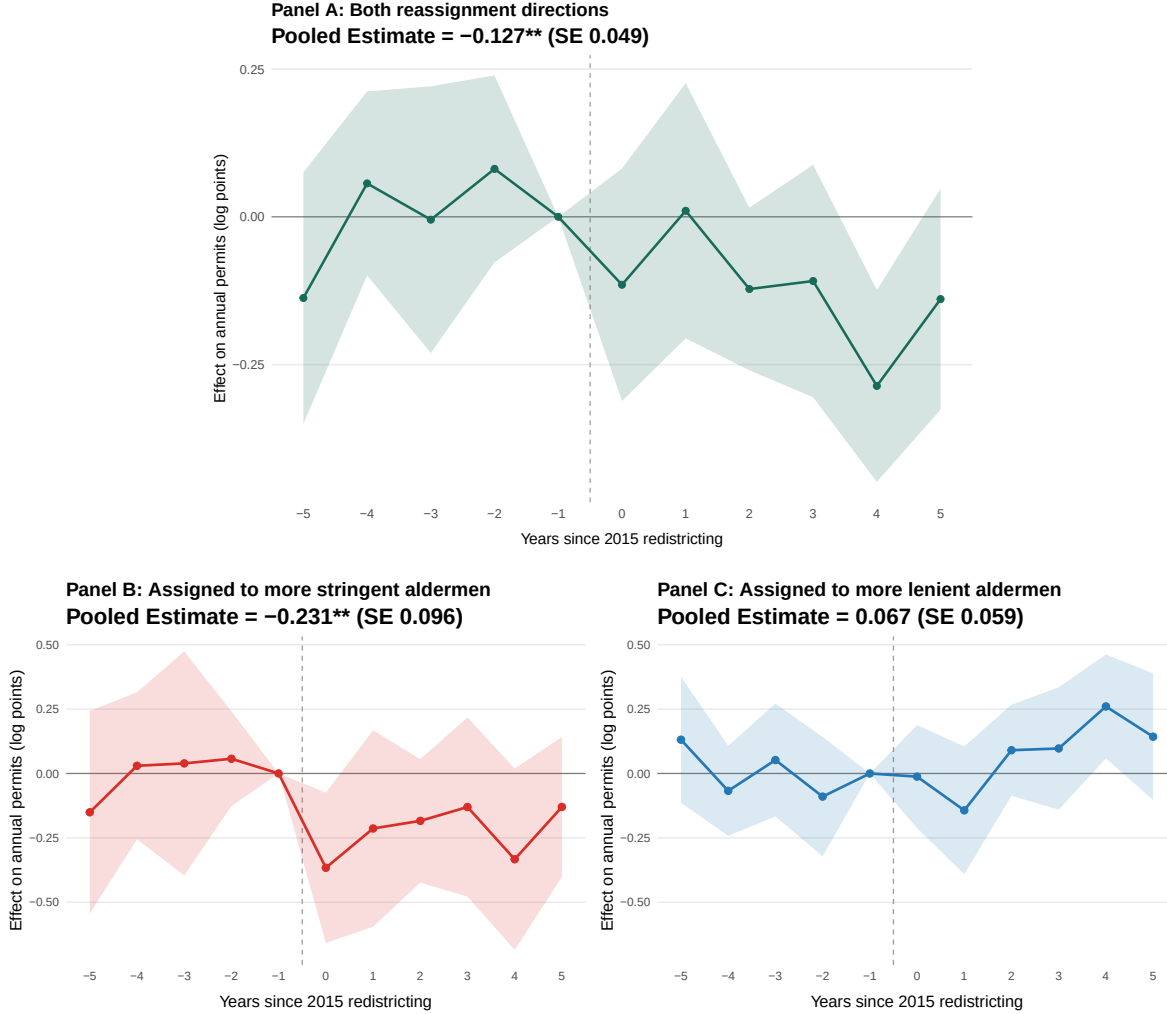


Figure 2: Permit Activity After the 2015 Ward Redistricting

Notes: The figure compares annual high-discretion permit activity on blocks reassigned to more- or less-stringent aldermen in 2015 with nearby blocks that stayed in their wards. The 2010–2020 sample includes blocks within 500 ft of ward boundaries; [Online Appendix A.3](#) details sample construction. Panel A combines both reassignment directions using Equation (3); Panels B and C allow the effects of the two directions to differ. Points show effects in log points relative to 2014; subtitles report pooled 2015–2020 coefficients on the same scale. Panel A’s pooled coefficient of -0.127 corresponds to an 11.9% reduction in annual permits for assignment toward greater stringency. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. The test that the pre-2015 effects in Panel A are jointly zero gives $p = 0.400$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

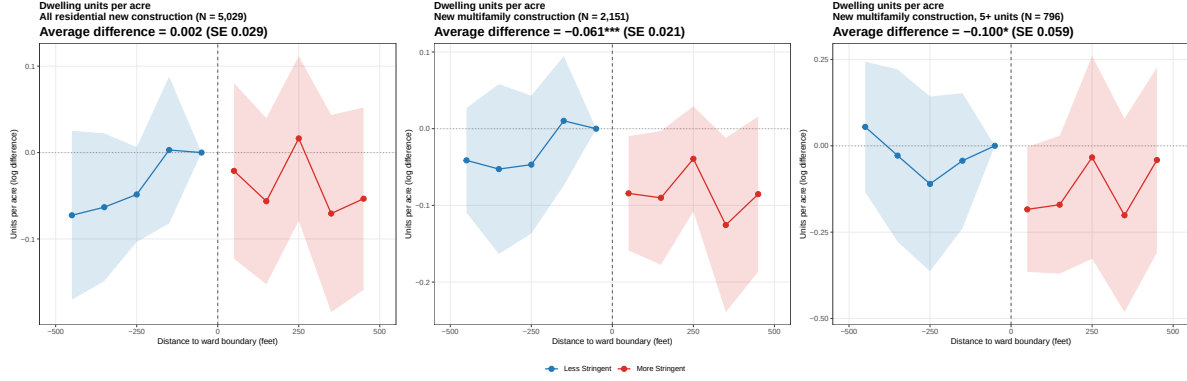


Figure 3: Development Density at Ward Boundaries

Notes: Panels show all residential new construction, new multifamily construction (two or more units), and new multifamily buildings with five or more units. The outcome is log dwelling units per acre. Samples include new buildings within 500 ft of ward boundaries with a recorded dwelling count and lot area. Each point is the coefficient for a 100 ft distance band from Equation (4). The nearest band on the less-stringent side, from -100 to 0 ft, is omitted and normalized to zero. Subtitles report the average difference between the two sides of the 500 ft window. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. Negative distance indicates the less-stringent side and positive distance the more-stringent side. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Listed rents and home sale prices at ward boundaries

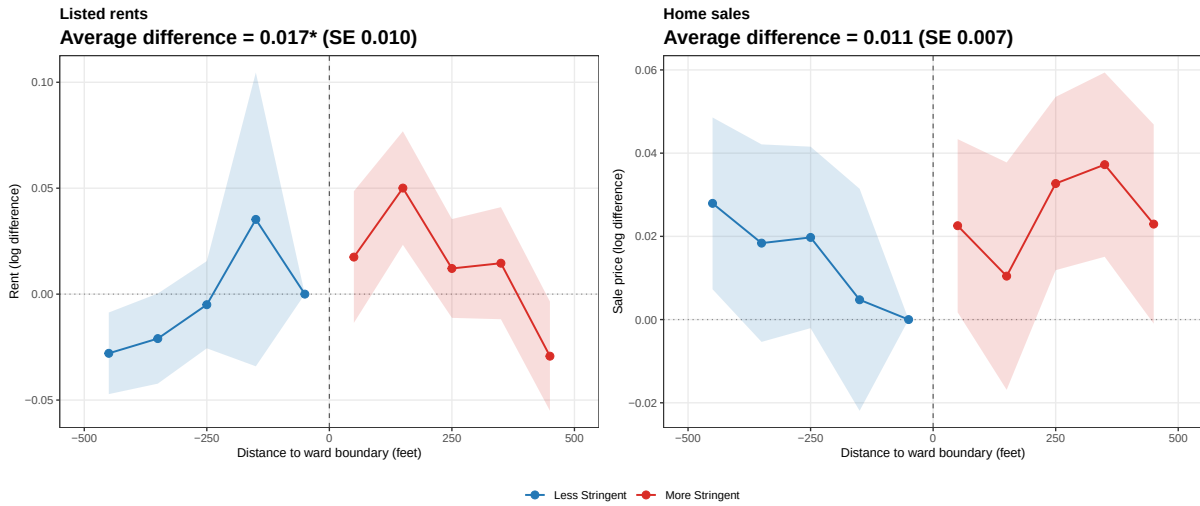


Figure 4: Listed Rents and Home Sale Prices at Ward Boundaries

Notes: The left panel uses 240,713 monthly asking rents by floorplan from 2014–2022, with segment-by-month fixed effects. The right panel uses 55,864 sales of detached houses, townhouses, and apartment buildings with two to six units from 2006–2022, with segment-by-quarter fixed effects. Both panels control for housing characteristics and distances to nearby amenities. Rental controls include listing building-type fixed effects; sales controls include Cook County Assessor property-class fixed effects. Each point is the estimated log difference between a 100 ft distance band and the band immediately inside the less-stringent side, which is normalized to zero. Shaded areas are 95% confidence intervals. Subtitles report the average difference between the two sides of the 500 ft window. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 1: New Residential Construction in Chicago, 2006–2022

	All New Construction	Multifamily	Multifamily, 5+ Units
Average Units per Building	5.97	15.31	36.16
Average DUPAC	30.75	55.65	83.06
Median Distance to Boundary (ft)	858	669	694
Ward Pairs / Boundary Segments	140/1446	116/972	94/661
Number of Buildings	15,350	5,322	1,977

Notes: One row per new residential building with a recorded dwelling count and land area, including condominium buildings; townhouses are counted one home at a time. DUPAC is dwelling units per acre. Multifamily buildings contain two or more dwelling units and are not single-family or townhouse homes.

A Data and Sample Construction

This appendix provides additional detail on the data sources and sample rules described in Section 2.

A.1 Data Sources

The property data come from Cook County Assessor sales, assessment, parcel, and improvement files. RentHub provides rental listings from 2014–2022. Building permits and ward boundaries come from the City of Chicago. The amenity controls use school locations, Chicago Park District boundaries, major streets, CTA stops, and Lake Michigan geometry; CTA distances use the permanent station network in place during the listing month or on the sale date.

Demographic controls come from American Community Survey block groups and 2010 and 2020 census blocks. Because block groups do not follow ward boundaries, I assign their population and household counts to wards in proportion to the area that falls inside each ward. I use the resulting counts to calculate ward-level tenure, race, ethnicity, and education shares. The ward income measure is the household-weighted average of block-group median household income. Small overlaps in historical ward maps are divided proportionally so that no part of a block group is counted twice.

A.2 High-Discretion Permits

The permit records cover Department of Buildings permits since 2006. The high-discretion group contains the permit types defined in Section 2.2; low-discretion placebo permits are the remaining routine types, excluding signs. The event-study outcomes omit records marked cancelled, revoked, or suspended.

Processing time is the number of days between application and issuance. The processing-time regression excludes missing dates, negative durations, and same-day permits. I assign

each permit to the ward containing its coordinates under the boundaries in force on its application date.

A.3 Permit Event-Study Panel

The treatment panel compares each census block’s ward before and after the 2015 redistricting. I assign a block to the ward covering the largest share of its area. Let $a_o(b)$ denote the alderman representing block b under the old map and $a_d(b)$ the alderman under the new map. Blocks moved to a different ward are classified using the difference between these aldermen’s stringency scores:

$$\Delta S_b = S_{a_d(b)} - S_{a_o(b)}.$$

Positive values indicate assignment toward greater stringency, negative values indicate assignment toward greater leniency, and unchanged blocks form the comparison group.

For this design, stringency scores use permits filed from 2006 through 2014. Both origin and destination scores are assigned using the incumbents serving in 2014. The main sample retains blocks only when both incumbents remained in office when the new map took effect, so treatment matches the change the block experienced without using the outcome of the 2015 election. Section C.4 adds wards with electoral turnover and interprets the resulting estimate as assigned rather than necessarily realized stringency.

I count permit applications by block and year from 2010 through 2020. The sample includes blocks whose geographic center lies within 500 ft of a pre-redistricting ward boundary. For reassigned blocks, distance is measured to the boundary separating the old and new wards; for unchanged blocks, it is measured to the nearest boundary of the original ward. The estimation sample requires at least one high-discretion permit during event times $t = -5$ through $t = -1$.

A.4 New Construction

I start from City of Chicago building permits issued between 2006 and 2022 for new dwellings, or for new buildings whose use the permit does not state. Most are new-construction permits, but some new buildings are filed as renovation, easy, or express permits, so I include those when the work description begins by erecting a new dwelling. Revisions, temporary structures, additions, conversions, garages and other accessory structures, and work on existing buildings are dropped. When the same address receives a second permit for the same number of dwellings within three years, I treat it as a repeat of the first.

I match each permit to the Cook County Assessor's residential improvement records, condominium records, and, from 2021, commercial apartment valuations. A record counts as a new building when its reported year built is between two years before and four years after the permit's issue year. Most permits are matched on their own parcels, meaning the parcels listed on the permit or the parcels at its address. When a parcel is later split into new lots or divided into condominium units, the permit is matched on the new parcel numbers that replaced it. A permit for several townhouses or houses that finds only some of its homes takes the adjacent new homes on the same block that complete its count. When a permit's parcels show no new building, it takes the unclaimed new building with a matching dwelling count within 150 ft of its geocoded point (300 ft for buildings of 20 or more units) that first appears after the permit, provided that building is the permit's only such match and the permit is the building's. Some rebuilt parcels keep their old year built in the Assessor records, so for a permit for a new residential building, a parcel within 250 ft of the permit also counts as the new building when its floor area rises by at least 25% within four years after the permit and stays at that level. A permit that still has no building takes the building on a parcel it lists that was created after the permit was issued and first assessed within four years of it, again when the two match only each other and their dwelling counts agree. Hand research, recorded with its evidence in the replication files, resolves matches these rules leave ambiguous, corrects matches found wrong in validation reviews, and measures condominium

buildings that the Assessor re-declared under several records. Permits matched to the same building are combined, so each building enters once. The permit file begins in January 2006, so buildings completed in 2006 and 2007 were often authorized by earlier permits; new buildings in the Assessor records reported built in those two years enter without a permit, while later buildings require one. Table [A.1](#) shows the results of this process.

The Assessor’s first record of a new building is often incomplete, since the building may still be under construction. I therefore use the characteristics a building’s records report most often in its first three assessment years. Condominium units are declared over several years, so for condominium buildings later years break ties. Each building is one observation, and townhouses and houses built under one permit enter as one observation per home, each measured on its own parcel. Condominium buildings are included.

Some records are not reliable enough to measure density. I drop a building’s density when the Assessor’s unit count does not match the permit (any difference for a single house, or for a multifamily building a difference of more than one unit or 20%, whichever is larger), when the records imply less than 300 sq ft of floor area per unit, more than an acre of land per unit, or less than 100 sq ft of land in total, or when an older building is still standing on the parcel. For buildings without a permit, I also drop density when another permit on the parcel describes work on an existing building, or when the first assessment comes more than four years after the reported year built. Large buildings sometimes report the land of a whole development rather than their own lot. For buildings of 20 or more units, I drop density when another large building within 1,000 ft reports the same land area, or when a building at least 80 ft tall covers less than a quarter of its land. Dwelling units per acre (DUPAC) is the number of dwelling units divided by lot acreage.

Buildings linked to permits are dated by their first permit’s issue date, and buildings without a permit by June 15 of their reported year built. Ward, neighboring-ward, and alderman assignments use the ward map and aldermen in office on that date. Each building is located at the centroid of its parcels, or at the permit’s geocoded point when no parcel

centroid is recorded. I calculate ward assignments and distances in the Illinois State Plane coordinate system used for Chicago (EPSG:3435) and report distances in feet.

Each building receives the broad zoning group in force when it was built. When the building’s parcel was last rezoned before construction, the group comes from the current (2025) zoning map. Otherwise it comes from the most recent official map before construction: a reconstruction of the 2006 map from passed map amendments for buildings dated through 2012, the 2012 map through 2014, the 2014 map in 2015, and the 2016 map afterward. Where the 2006 reconstruction has no polygon, the 2012 map applies when the 2012, 2014, and 2016 maps agree. Planned developments form a separate zoning group.

A.5 New-Construction Summary Statistics

Table 1 describes the citywide sample of 15,350 buildings completed from 2006 through 2022 with a recorded dwelling count and lot area, including 5,322 multifamily buildings and 1,977 buildings with five or more units. Of these, 3,404 are condominium buildings and 2,141 are buildings reported built in 2006 or 2007 without a matching permit.

A.6 Home Sales

The raw file contains 902,830 parcel-sale records in Chicago’s eight townships from 2006–2022. The transaction sample includes Assessor classes 202–210, 234, 278, and 295, which cover detached houses and townhouses, and class 211, which covers apartment buildings with two to six units. I exclude mixed-use class 212, condominiums, cooperatives, larger apartment buildings, residential land, and all other property classes.

Transactions must be market sales: a nominal price above \$10,000, not explicitly identified as land-only, one parcel, and passing the three sale filters the Assessor applied to its published sales data before October 2023, which remove sales at the same price as another sale of the same property within 365 days, sales below \$10,000, and quit claim, executor, beneficiary, and missing deed types. Of the 296,779 such sales, I remove 6,405 foreclosure

auctions and 1,208 transfers to a lender, federal agency, or land bank, identified by party names and, for sales without a seller name, by the deed type recorded from 2013 onward. I also remove 47,824 resales of foreclosed homes by lenders, loan servicers, securitization trustees, federal housing agencies, or the land bank, whose prices reflect the distress of the sale rather than the location. Finally, I keep only warranty and trustee deeds, removing 14,067 further sales recorded under special warranty, administrator's, or other deeds, many of them foreclosure resales whose sellers the party names do not identify. These rules leave 227,275 sales with valid months and full 14-digit PINs.

Some older records without an IDOR-refined date report the first day of the month even though the exact day is unavailable. I exclude these sales when either adjoining ward changes aldermen during that month, or when the ward map changes during the month, because the record does not identify the relevant alderman or ward map. This removes 1,194 transactions citywide and leaves 226,081.

I measure property characteristics using the Assessor record for the same property and tax year as the sale. I keep properties with one recorded building, positive building area, and an Assessor proration rate of one, which indicates that the record describes the entire property rather than a fractional share. These restrictions leave 222,549 sales. I then exclude 40 sales whose property records report more bedrooms than total rooms and 4 sales priced above \$5,000 per square foot of building area, which indicate recording errors, leaving 222,505 sales. When the sale and improvement files report different residential subclasses, I use the class recorded in the sale file for the property-class fixed effect. I express prices in 2022 dollars using the monthly Chicago CPI-U, with the mean 2022 index as the base.

I locate each transaction by matching its PIN to the parcel map for the year of sale; all 227,275 sales meeting the transaction-quality criteria receive an exact historical location. For retained transactions, ward, neighboring ward, and alderman assignments use the recorded sale date: transactions before May 18, 2015, use the 2003 ward map, and those on or after that date use the 2015 map. I calculate distance to the nearest boundary of the

property’s ward in the Illinois State Plane coordinate system. Distance is positive on the side represented by the more-stringent alderman and negative on the other side. I drop boundaries if either alderman lacks a stringency score or the two scores are identical.

For the boundary-price design, I split ward-pair boundaries into approximately 1,320-ft segments and assign each sale within 1,500 ft to the nearest segment on its ward-pair boundary. To compare similar homes in similar locations, the regression controls for log building area, log land area, log building age, log bedrooms, log bathrooms, a garage indicator, and distances to the nearest school, park, major road, CTA station open on the sale date, and Lake Michigan. Building age is sale year minus reported year built, and total bathrooms equal full bathrooms plus one-half of half bathrooms. The logged property characteristics must be positive, so properties sold in their reported construction year are excluded because their measured age is zero. The preferred 500-ft sample contains 55,864 sales. Property-class fixed effects compare sales within the same Assessor class, while segment-by-sale-quarter fixed effects compare sales near the same short stretch of boundary in the same three-month period. Standard errors are clustered by ward pair.

A.7 RentHub Listings

The rental data record each occasion on which RentHub observed a listing. I standardize addresses and building types and treat blank or placeholder addresses as missing. Because RentHub does not report a consistent unit identifier, I group observations into properties using the standardized address and rounded coordinates, or coordinates alone when the address is missing. Within each property, I identify a floorplan by its bedrooms, bathrooms, square footage, and building type. I take the median asking rent when the same floorplan appears more than once on a date and then take the median across dates within each month. Before monthly aggregation, rent and square footage are trimmed at the 1st and 99th percentiles within year-by-bedroom cells.

The sample requires positive rent, valid coordinates, and assignment to a Chicago ward.

Ward and alderman assignments use the listing's first observed date, with May 18, 2015, as the map transition date. The analysis excludes listings with coordinates that fail manual review, addresses that appear at inconsistent locations, and addressless listings concentrated at generic map points. It also excludes an address when replacing a listing's coordinate with that address's most frequently recorded location changes its ward assignment or moves its measured boundary distance by more than 100 ft. Regressions also require square footage, bathrooms, and amenity controls. Bedrooms enter categorically, so studios remain in the sample.

Table A.1: How New Buildings Are Matched to Permits

	All New Construction	Multifamily	Multifamily, 5+ Units
Permit's own parcels	9,555	2,434	1,025
Successor parcels	2,849	1,573	419
Townhouse rule	209	0	0
Nearby-lot rule	172	116	83
Floor-area increase	127	88	13
Parcel created after the permit	19	17	8
Hand research	278	25	8
No permit (built 2006–2007)	2,141	1,069	421
Total	15,350	5,322	1,977

Notes: Counts of the buildings in Table 1 by the step in Section A.4 that matched each building to its permit. Townhouses are counted one home at a time; all homes on a permit that the townhouse rule completed are counted under that rule.

B Alderman Fixed Effects and Index Validation

This appendix provides additional detail on how the aldermanic stringency index is constructed and validated.

B.1 Construction of the Stringency Index

Aldermanic stringency measures whether comparable high-discretion permits take systematically longer under a particular alderman. It uses the permit sample described in Section A.2 through December 2022. Permits must take at least one day to enter the processing-time comparison, although same-day permits still count toward the ward’s recent permit volume.

Stage 1: Adjusting Processing Times. I first adjust processing times for observable reasons that otherwise could make permits slower in some wards. The regression controls for the permit’s distance to downtown and Lake Michigan, nearby CTA stations, ward demographics, recent ward permit volume, year-month, permit type, and review type. Appendix A describes how block group demographics are allocated across ward boundaries to construct the ward controls.

Stage 2: Alderman Differences. I average the adjusted processing times by ward and month, then compare these averages across the aldermen serving each ward-month. Formally, I regress the ward-month averages on alderman indicators $\alpha_{a(w,m)}$ and the same recent permit-volume control. Ward-months are weighted by their permit counts. Here $a(w, m)$ denotes the alderman serving ward w in month m . The resulting fixed effect is the persistent alderman-specific difference in adjusted processing time.

Empirical Bayes Shrinkage. Some aldermen have fewer permits and therefore less precise estimates. I use empirical Bayes shrinkage to pull these noisier estimates toward the overall mean while leaving more precise estimates closer to their raw values: $\hat{\alpha}_a^{\text{EB}} = B_a \hat{\alpha}_a$,

where $B_a = \hat{\omega}^2 / (\hat{\omega}^2 + \hat{\sigma}_a^2)$. Here $\hat{\omega}^2$ is the estimated variance across aldermen and $\hat{\sigma}_a^2$ is the sampling variance of alderman a 's estimate.

Index Standardization. The stringency index, S_a , is standardized to have mean zero and unit standard deviation across aldermen. Higher values indicate more stringent aldermen.

Table B.1 reports the adjustment regression used in the first stage. The dependent variable is permit-level log processing time. These controls remove predictable differences in timing across locations and permit types before I compare aldermen. Panel B of Figure B.1 shows the distribution of the stringency scores.

B.2 Validation of the Stringency Index

The stringency index should measure differences across aldermen rather than permanent differences across wards. Panel C of Figure B.1 compares predecessor and successor aldermen within the same ward. If fixed ward characteristics drove the scores, aldermen serving the same ward at different times would receive similar values. The relationship is weak, which weighs against that explanation.

Aldermen's public statements on prerogative provide a second check. I collected 76 statements by 63 scored aldermen, most of them answers to the 2019 *Chicago Sun-Times* candidate questionnaire, and coded whether each supports keeping, reforming, or abolishing prerogative. Table B.2 shows that aldermen who publicly supported reforming or abolishing prerogative have lower stringency scores, and those who supported keeping it have higher scores. Because most statements date from 2019 to 2023, the relationship comes mainly from aldermen who took office after 2014.

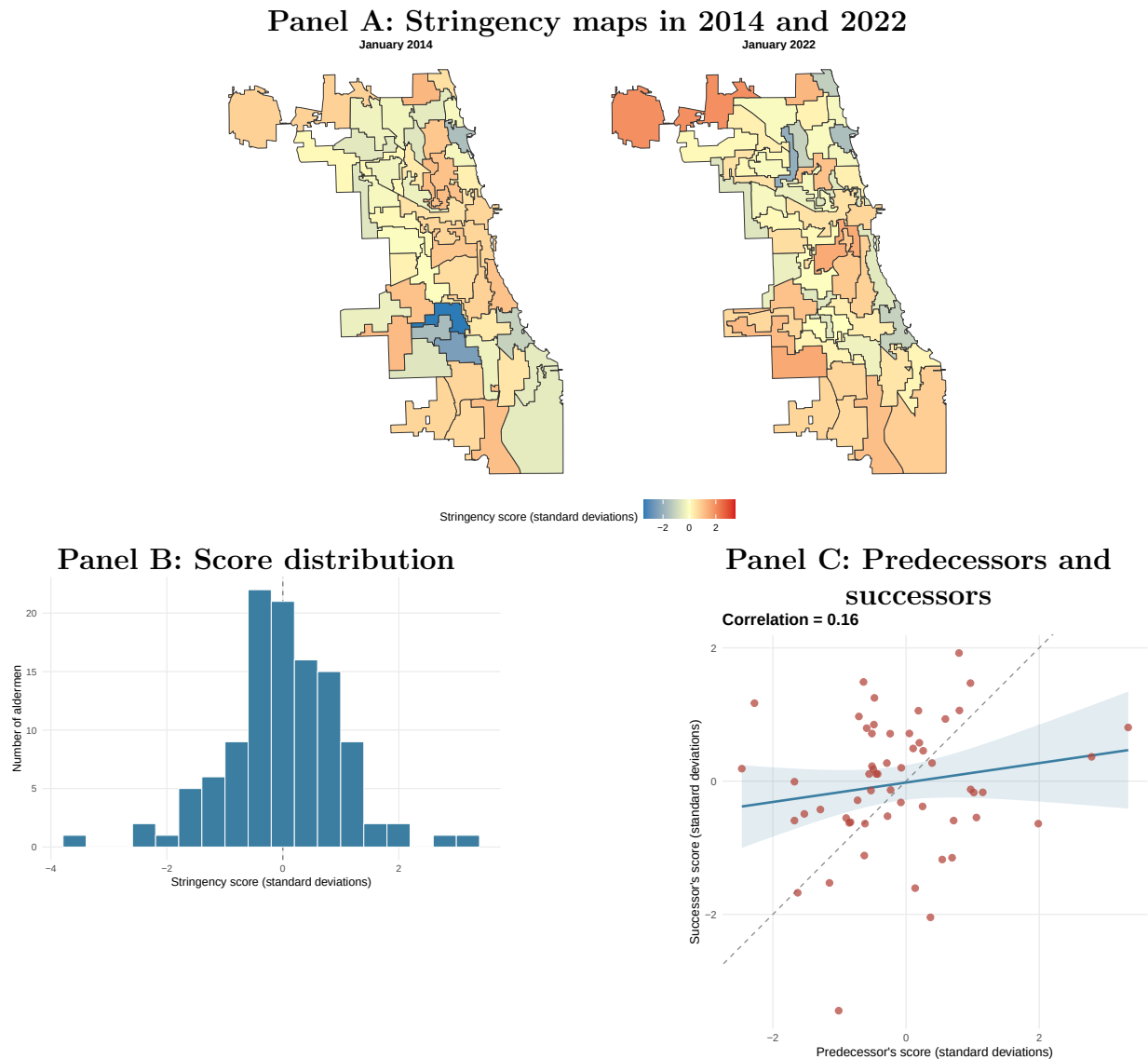


Figure B.1: Variation in Aldermanic Stringency

Notes: Stringency scores use permit processing times through December 2022 after adjusting for permit, location, ward, and calendar characteristics. Noisier alderman estimates are pulled toward the overall mean, and the resulting scores are standardized to mean zero and unit standard deviation. Panel A maps the scores of aldermen serving in January 2014 and January 2022 onto the corresponding ward boundaries; both maps use the same color scale, with warmer shades indicating more stringent aldermen. Panel B's dashed line marks the mean score of zero. Each point in Panel C pairs successive aldermen from the same ward. The dashed line indicates equal predecessor and successor scores; the solid line is a linear fit, with a shaded 95% confidence interval for the fitted mean.

Table B.1: Factors Associated with Permit Processing Times

	Log Processing Time (days) (1)
Median Household Income (\$10,000)	-0.015 (0.016)
Black Population Share	0.176 (0.278)
Hispanic Population Share	0.404 (0.272)
White Population Share	0.530 (0.352)
Homeownership Rate	0.261 (0.222)
Population (10,000)	-0.002 (0.020)
Distance to Downtown (km)	-0.017*** (0.006)
Distance to Lake (km)	-0.002 (0.005)
CTA Stations within 800 m	-0.009*** (0.003)
Permits in Previous Month	-0.002*** (0.0006)
Year-Month Fixed Effects	✓
Permit Type Fixed Effects	✓
Review Type Fixed Effects	✓
N	120,505

Notes: Permit-level linear regression used to construct the stringency index, using permit records through December 2022. Covariates are median household income, racial shares, homeownership, population, distance to downtown, distance to the lake, nearby CTA stations, and the previous month's permit volume. Fixed effects are listed in the table. Regressions are unweighted. Standard errors are clustered by ward and reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.2: Stringency Scores and Stated Positions on Aldermanic Prerogative

	Stringency score	
	(1)	(2)
Stated support for keeping prerogative	0.349* (0.192)	
Stated support for reforming or abolishing prerogative		-0.616*** (0.221)
Constant	0.045 (0.135)	0.385*** (0.101)
N	63	63

Notes: Each column regresses the 2006–2022 stringency score on an indicator for the alderman’s stated position, with one observation per alderman with at least one public statement. Statements come from the 2019 *Chicago Sun-Times* candidate questionnaire, the 2019 WBEZ mayoral questionnaire, the 2023 WTTW voters’ guides, and news coverage, and each quote was checked against its source. Aldermen with mixed or unclear statements are in the comparison group of both columns. Heteroskedasticity-robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Additional Permit Evidence

This appendix reports additional permit evidence that complements the descriptive discussion in Section 2 and the event study in Section 4.1.

C.1 Balance Before Redistricting

Table C.1 compares blocks that changed wards in 2015 with nearby blocks that remained in the same ward. Before the new map took effect, the two groups were broadly similar in permit activity and neighborhood characteristics. The joint test across all listed characteristics has a p -value of 0.333.

C.2 Permit Event-Study Estimates

Table C.2 summarizes the average change in annual permit counts during the first six years under the new ward map. The combined model treats reassignment toward greater stringency and greater leniency as changes of equal size in opposite directions. A second model allows the two directions to have different effects. Low-discretion permits show a more muted response. In the main models, unchanged blocks on both sides of the old boundary between the origin and destination wards form the comparison group. The last block compares each reassigned block only with unchanged blocks in the ward it left, which shared its alderman before 2015 and kept that alderman afterward. This comparison gives similar high-discretion estimates, -0.132 for the combined effect and -0.261 for moves toward greater stringency, with larger standard errors.

C.3 Allowing the Two Reassignment Directions to Differ

Panels B and C of Figure 2 allow moves toward greater stringency and greater leniency to have different effects. Their difference measures how permit activity changes with stringency, and Table C.2 shows that it is similar to the combined estimate.

Panel A of Figure C.1 shows the corresponding path for low-discretion permits. The response is more muted than for high-discretion permits.

C.4 Including Wards with Alderman Turnover

Panel B includes blocks from wards that elected new aldermen in 2015. The reassignment classification still uses the two aldermen serving in 2014. For blocks whose alderman subsequently changed, this measures the change implied by the map rather than the stringency the block necessarily experienced under the newly elected alderman. The annual number of high-discretion permits is 8.3% lower, a somewhat smaller estimate than in the main sample.

C.5 A Symmetric Window Before the 2019 Elections

Many blocks' wards elected new aldermen in the 2019 elections, four years after the new map took effect. Figure C.2 repeats the event study in Figure 2 on a symmetric window from 2012 through 2018, three years on either side of the redistricting. The sample still requires at least one high-discretion permit during 2010–2014. The estimates are very similar to those for the full window: -0.114 for the combined effect, -0.257 for moves toward more stringent aldermen, and 0.031 for moves toward more lenient aldermen, compared with -0.127 , -0.231 , and 0.067 .

C.6 Continuous Stringency Changes

Equation (3) uses only the direction of each move. Figure C.3 replaces $Z_b = \text{sign}(\Delta S_b)$ with the score change ΔS_b itself, in standard deviations of the 2006–2014 score, and zero for blocks that remain in the same ward. Each coefficient is then the effect of a one-standard-deviation increase in assigned stringency. The pooled 2015–2020 estimate is -0.143 ($\text{SE} = 0.043$), a 13.4% reduction in high-discretion permits per standard deviation, and the test that the pre-2015 effects are jointly zero gives $p = 0.765$.

C.7 Low-Discretion Permits as an Additional Comparison

Low-discretion permits on the same blocks provide a second comparison group, because shocks that affect all permitting on a block, such as local economic conditions, should move both groups. I stack high- and low-discretion permit counts by block b , permit group c , and year y and estimate

$$\log \mathbb{E}[Y_{bcy} \mid \cdot] = \mu_{bc} + \lambda_{g(b),c,y} + \sum_{t \neq -1} \beta_t Z_b \mathbf{1}\{y - 2015 = t\} + \sum_{t \neq -1} \gamma_t Z_b H_c \mathbf{1}\{y - 2015 = t\}, \quad (\text{C.1})$$

where H_c indicates high-discretion permits, μ_{bc} are block-by-group fixed effects, and $\lambda_{g(b),c,y}$ are ward-pair-by-year-by-group fixed effects. The coefficients γ_t measure how much more high-discretion permits change than low-discretion permits on the same blocks. Figure C.4 plots them for the combined model and, with separate indicators, for each direction. Relative to low-discretion permits, the combined effect is -0.109 ($\text{SE} = 0.058$); moves toward more stringent aldermen reduce high-discretion permits by 0.269 log points ($\text{SE} = 0.111$), and the estimate for moves toward more lenient aldermen is small and statistically insignificant (0.024 , $\text{SE} = 0.065$). The tests that the pre-2015 coefficients are jointly zero give $p = 0.965$, 0.653 , and 0.226 .

C.8 Permit Summary Statistics and Volume

Table C.3 compares the high- and low-discretion permit groups and reports their ward-month processing-time and permit-volume correlations. High-discretion permits are much slower than low-discretion permits: the mean processing time is 57.3 days versus 26.2 days, and the median is 30 days versus 6 days. They also display more cross-alderman variation in processing times. Ward-months with longer high-discretion processing times have fewer high-discretion permits. Low-discretion permits show a weak positive relationship instead.

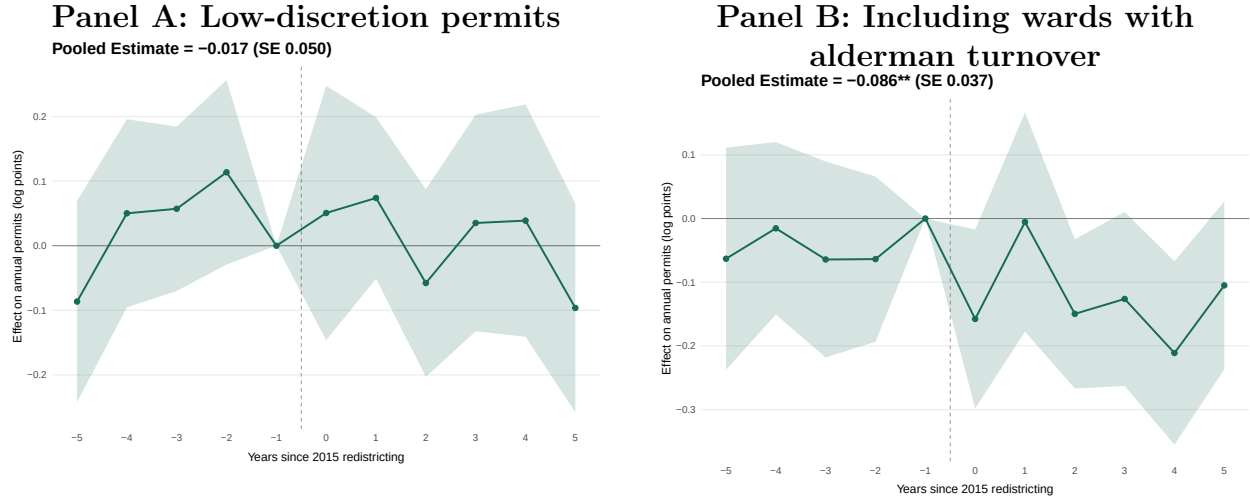


Figure C.1: Additional Permit Event-Study Checks

Notes: Panel A uses low-discretion permits excluding signs. Panel B uses high-discretion permits and includes wards with alderman turnover; assignment still uses the 2014 incumbents. Both panels combine the two reassignment directions using Equation (3); negative coefficients indicate fewer permits under assignment toward greater stringency. Points show effects in log points relative to 2014; subtitles report pooled 2015–2020 coefficients on the same scale. The pooled effects correspond to reductions of 1.7% and 8.3% in annual permits, respectively. Tests that the pre-2015 effects are jointly zero give $p = 0.075$ in Panel A and $p = 0.666$ in Panel B. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

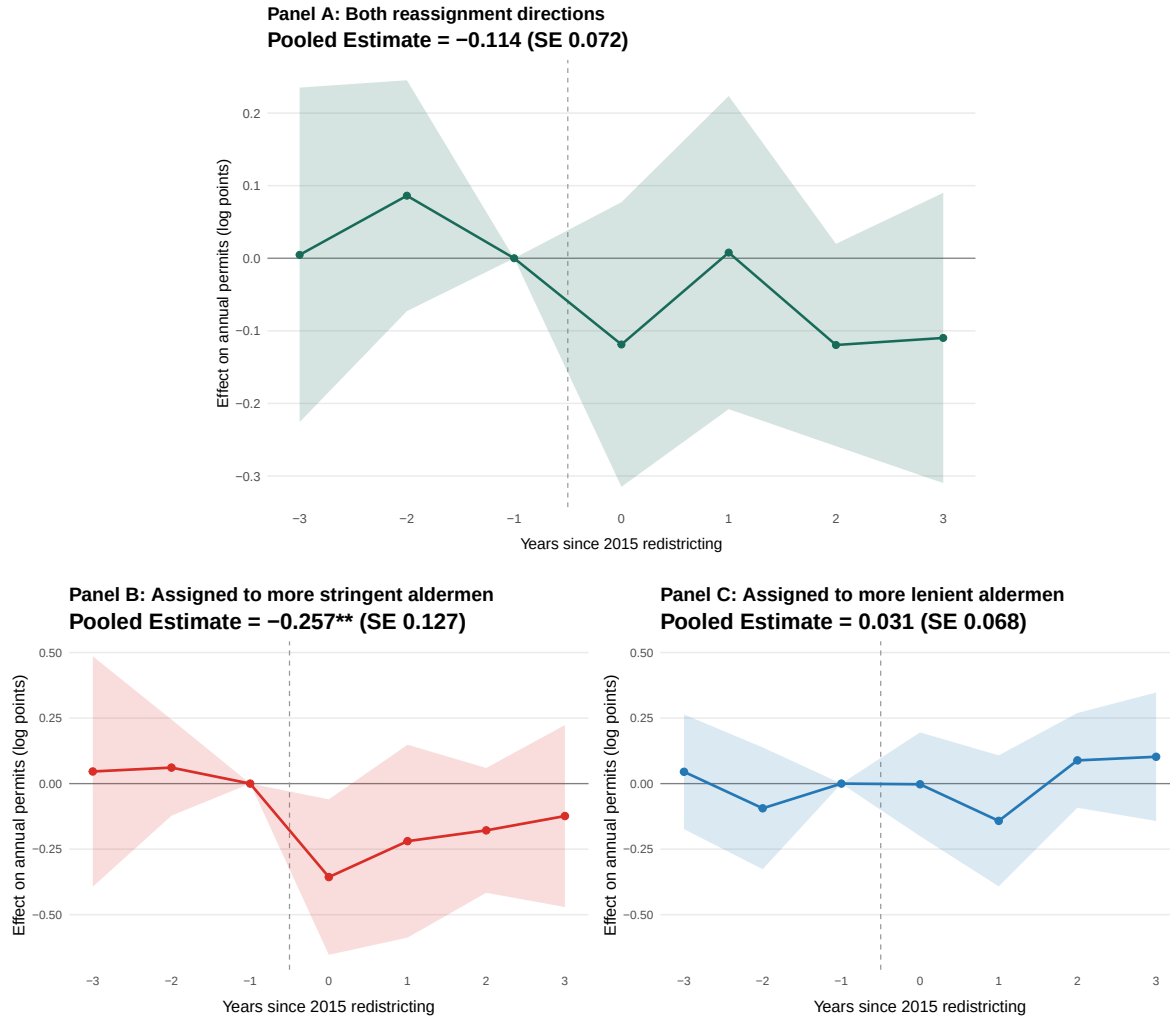


Figure C.2: Permit Activity After the 2015 Ward Redistricting, 2012–2018

Notes: The figure repeats Figure 2 on the 2012–2018 window, before the 2019 elections. The sample includes blocks within 500 ft of ward boundaries with at least one high-discretion permit during 2010–2014. Panel A combines both reassignment directions using Equation (3); Panels B and C allow the effects of the two directions to differ. Points show effects in log points relative to 2014; subtitles report pooled 2015–2018 coefficients on the same scale. The test that the pre-2015 effects in Panel A are jointly zero gives $p = 0.551$. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

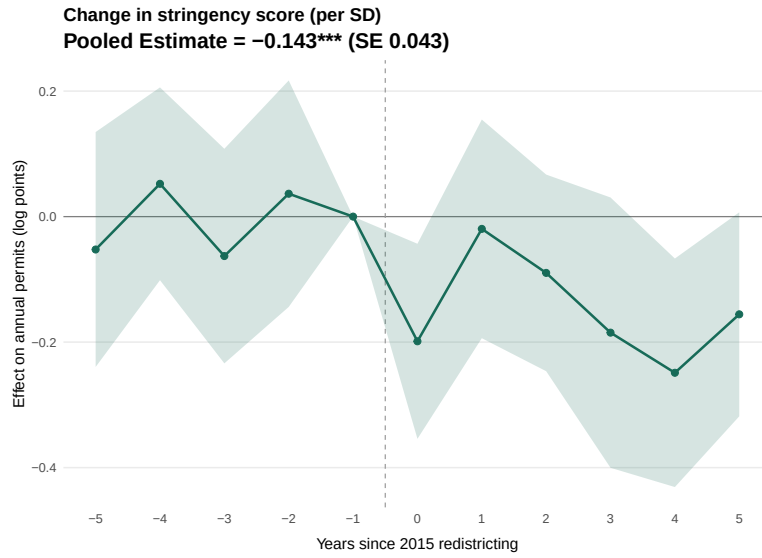


Figure C.3: Permit Activity After the 2015 Ward Redistricting, Continuous Stringency Change

Notes: The figure repeats Panel A of Figure 2 with the destination-minus-origin score change ΔS_b in place of its sign. Points show effects in log points per standard deviation of score change, relative to 2014; the subtitle reports the pooled 2015–2020 coefficient. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

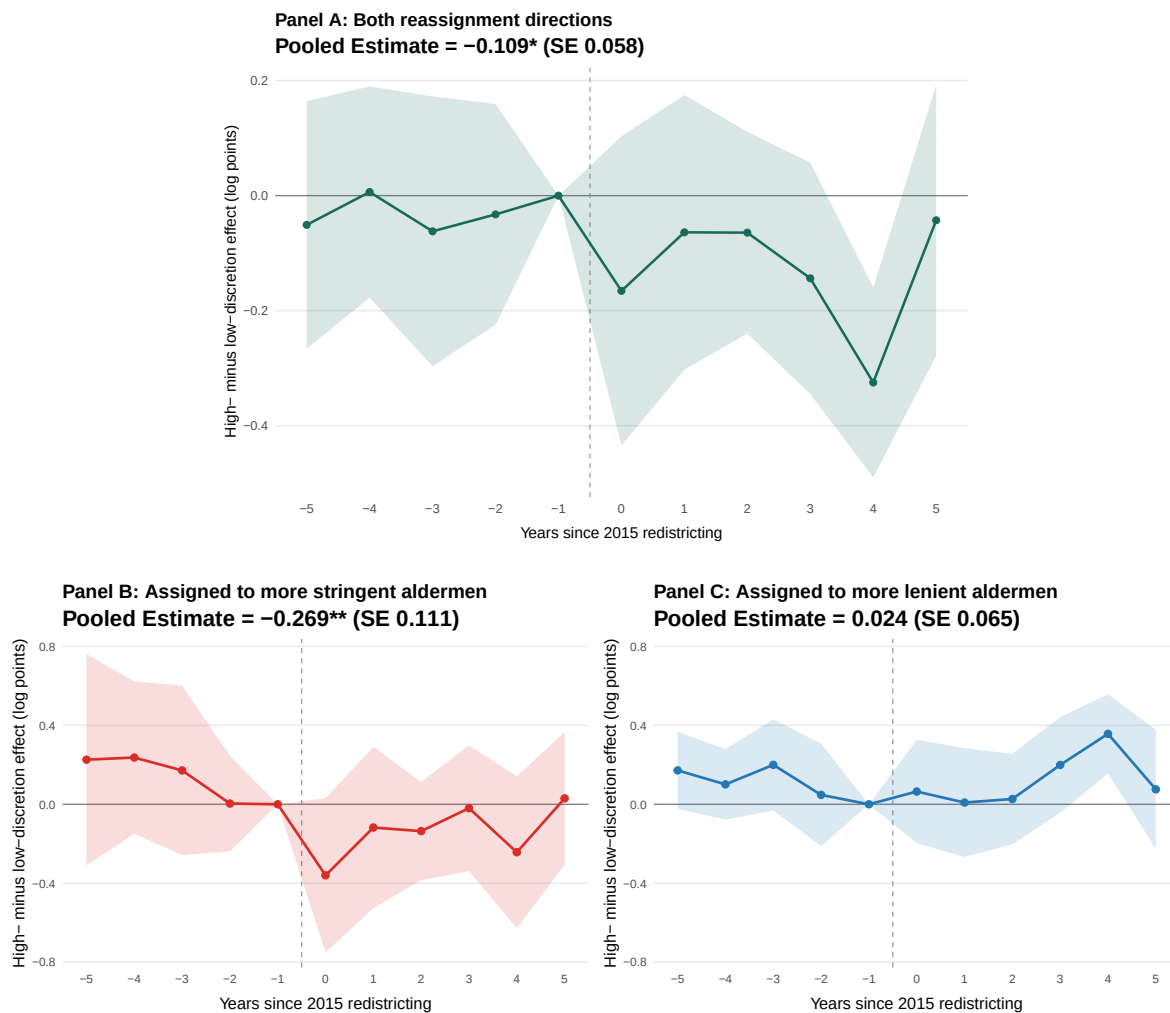


Figure C.4: High- Relative to Low-Discretion Permit Activity After the 2015 Ward Redistricting

Notes: Points show the coefficients γ_t from Equation (C.1), in log points relative to 2014: the change in high-discretion permits beyond the change in low-discretion permits (excluding signs) on the same blocks. Panel A combines both reassignment directions; Panels B and C allow them to differ. The sample matches Figure 2; subtitles report pooled 2015–2020 coefficients. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.1: Pre-Redistricting Balance Between Reassigned and Unchanged Blocks

Variable	Unchanged	Reassigned	Difference	SE	<i>p</i> -value	<i>N</i>
<i>Permit history</i>						
High-discretion permits, 2010–2014	2.668	2.649	0.086	0.081	0.291	2,340
Low-discretion permits, 2010–2014	4.278	4.377	0.146	0.412	0.725	2,340
No low-discretion permit, 2010–2014	0.114	0.127	0.008	0.023	0.713	2,340
New-construction permits, 2010–2014	0.381	0.349	0.013	0.044	0.766	2,340
No new-construction permit, 2010–2014	0.773	0.798	-0.008	0.013	0.525	2,340
<i>Neighborhood characteristics</i>						
Median household income	58,210	53,228	-1,352	1,354	0.324	2,324
Homeownership rate	0.404	0.407	0.000	0.011	0.998	2,340
Bachelor’s degree or higher share	0.441	0.399	-0.000	0.009	0.970	2,340
Black population share	0.301	0.272	0.003	0.006	0.667	2,340
Hispanic population share	0.226	0.276	-0.011	0.009	0.230	2,340
Median gross rent	1,068	1,022	-19	16	0.246	2,285
Median home value	309,400	283,787	-8,561	11,019	0.441	2,288
Average household size	2.550	2.694	-0.004	0.052	0.932	2,340
Median age	33.8	33.5	0.1	0.4	0.798	2,340
Population per square kilometer	8,350	8,187	-100	378	0.791	2,340

Notes: Reassigned blocks moved to a different ward under the 2015 map; unchanged blocks remained in their original ward. The sample requires the 2014 origin- and destination-ward incumbents to remain in office when the new map took effect. It also requires at least one high-discretion permit during 2010–2014. The sample contains 687 reassigned and 1,653 unchanged blocks within 500ft of a ward boundary in 44 ward pairs containing both groups. Permits are grouped by application year. Neighborhood characteristics are 2014 ACS five-year block-group estimates. Difference compares reassigned and unchanged blocks within the same ward pair; standard errors are clustered by ward pair. Means are unadjusted block-level means. The joint-test *p*-value for all listed covariates is 0.333 (complete-case $N = 2,249$).

Table C.2: Permit Effects with Combined and Separate Reassignment Directions

	High-Discretion	Low-Discretion
Combined reassignment effect	-0.127** (0.049)	-0.017 (0.050)
<i>Directions estimated separately</i>		
Assigned to more stringent aldermen	-0.231** (0.096)	0.037 (0.074)
Assigned to more lenient aldermen	0.067 (0.059)	0.043 (0.065)
One-half difference between directions	-0.149*** (0.052)	-0.003 (0.048)
Equal-and-opposite test p -value	0.180	0.432
<i>Comparison blocks from the original ward only</i>		
Combined reassignment effect	-0.132** (0.066)	-0.071 (0.069)
Assigned to more stringent aldermen	-0.261** (0.126)	-0.077 (0.090)
Assigned to more lenient aldermen	0.049 (0.073)	0.067 (0.092)
Block fixed effects	Yes	Yes
Ward-pair $\times$ year fixed effects	Yes	Yes
Positive pre-period permit activity required	Yes	Yes
N	36,830	35,541

Notes: Poisson regressions use census-block-year observations from 2010–2020 within 500 ft of a ward boundary. Stringency scores are estimated using permits filed from 2006 through 2014. The sample retains blocks whose 2014 origin- and destination-ward incumbents remained in office when the new map took effect. It also requires at least one high-discretion permit during event times $t = -5$ through $t = -1$. Blocks are classified as assigned toward greater stringency, assigned toward greater leniency, or unchanged under the new map. The combined model imposes equal-and-opposite effects for the two reassignment directions, with unchanged blocks as the reference group; a negative coefficient means that greater stringency reduces permits. The separate model includes one indicator for each reassignment direction. Its half-difference row puts the two-direction estimates on the same scale as the combined estimate. The final test asks whether the separately estimated effects are equal in size and opposite in sign. Each model includes block fixed effects and ward-pair $\times$ year fixed effects, except that the last block uses ward-pair $\times$ original-ward $\times$ year fixed effects so that reassigned blocks are compared only with unchanged blocks in the ward they left. The estimator omits blocks with no permits of the relevant type in any sample year and ward-pair-year groups with no such permits. Standard errors, in parentheses, are clustered by ward pair.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Processing-time summary statistics

Statistic	High-Discretion	Low-Discretion
Total permits in sample	121,405	124,124
Mean processing time (days)	57.3	26.2
Median processing time (days)	30.0	6.0
Alderman-level mean processing time (days)	60.2	26.6
Alderman-level IQR of mean processing time (days)	24.5	13.5
Number of unique wards	50	50
Number of unique aldermen	114	114

Panel B: Ward-month processing time and permit volume

Permit group	Raw correlation	Elasticity within ward and month
High-Discretion	-0.123	-0.042** (0.019)
Low-Discretion	0.106	0.077*** (0.021)
All	-0.038	-0.038* (0.022)

Table C.3: Summary Statistics for High- and Low-Discretion Permits

Notes: The sample uses permits issued at least one day after application, with applications begun through December 2022. The permit groups are defined in Section 2.2. The alderman-level interquartile range is computed across alderman-specific mean processing times. In Panel B, each observation represents one ward in one month. The raw correlation is the Pearson correlation between mean processing time and permit volume. The elasticity is the coefficient from a regression of mean log processing time on log permit volume with ward and year-month fixed effects; standard errors are clustered by ward. The All row pools high- and low-discretion permits. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Additional Density Boundary Checks

This appendix asks whether the density result could be an artifact of how I measure aldermanic stringency or where ward boundaries happen to lie. It also tests for jumps away from true boundaries and checks whether buildings immediately next to a boundary drive the result. Unless otherwise noted, each check uses Equation (4): the outcome is log dwelling units per acre, and the model includes boundary-segment-by-joint-service fixed effects, zoning-group fixed effects, and the ward demographic controls. A joint-service period is an uninterrupted interval during which the same two aldermen represent the two wards of a boundary. Each check reports two estimates: the difference between the two 100 ft bands immediately across the boundary, and the average difference, which replaces the distance bands with an indicator for the more-stringent side of the 500 ft window. All checks use buildings with a recorded dwelling count and lot area.

D.1 Stringency-Score Checks

The first check asks whether a building’s own permits influence the score used to classify its boundary. For each building, I recalculate the scores for the two aldermen after removing the permits matched to that building. Each building contributes at most a few of the more than 100,000 permits behind the scores, and the estimates are unchanged at three decimal places. The second check drops boundaries where the two aldermen receive similar scores and the more-stringent side is therefore harder to distinguish. The multifamily estimates remain negative when the required score difference is 0.25 or 0.50 standard deviations, but they are smaller and no longer statistically significant as the sample shrinks.

D.2 Placebo Cutoffs

Figure D.1 moves the cutoff 1,000 ft inside either side of the true ward boundary. These locations do not mark a change in aldermanic authority. None of the twelve placebo estimates

(two cutoffs, three samples, and two estimates each) is statistically distinguishable from zero at the 10% level. At the placebo cutoffs, the average difference for multifamily buildings is -0.022 and -0.030 , compared with -0.061 at the true boundary.

D.3 Continuity in Location and Neighborhood Characteristics

Table D.2 tests whether fixed features of a building’s location, and the census block around it, change at the ward boundary. Of the twelve differences in distance to downtown, the nearest school, the nearest park, and Lake Michigan, one is statistically significant, with multifamily buildings on the more-stringent side 0.010 miles closer to Lake Michigan. The second panel uses the 2000 census block containing each building, measured before any building in the sample was permitted. Blocks on the two sides have similar homeownership, vacancy, and housing density. Among all new construction, blocks immediately on the more-stringent side have a lower non-Hispanic white share (5.3 percentage points) and a slightly lower non-Hispanic Black share (1.2 points); for multifamily buildings neither difference is statistically significant.

D.4 Boundary Geometry

Some ward boundaries follow expressways, waterways, arterial roads, or other physical features. Table D.3 therefore repeats the analysis after restricting the sample based on the geometry of the assigned boundary segment. Removing segments with substantial expressway or water overlap leaves the estimates almost unchanged. The broader physical-feature and arterial-road restriction reduces the multifamily average difference to -0.037 , while the estimates for buildings with five or more units become larger. Restricting the sample to locally straight boundaries leaves negative multifamily estimates of similar size. For all construction, the estimates remain close to zero under every restriction. The last block replaces the segment-by-joint-service fixed effects with segment-by-construction-year fixed effects, which compare buildings on the same segment completed in the same year regardless

of which aldermen served. The multifamily average difference is larger, -0.105 compared with -0.061 , and the all-construction estimate remains close to zero.

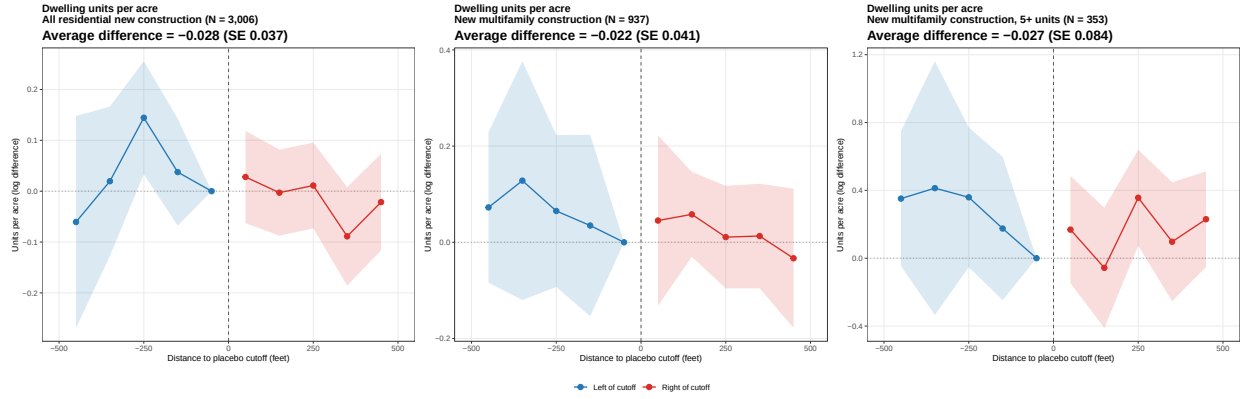
D.5 Excluding Buildings Nearest the Boundary

Figure D.2 drops buildings within 25 ft or 50 ft of the boundary. The multifamily estimates remain negative and slightly larger than in the full sample, and the all-construction estimates remain close to zero.

D.6 Continuous Stringency Differences

Figure D.3 multiplies the distance bands in Equation (4), and the more-stringent-side indicator behind the average difference, by the absolute score gap $|S_a - S_{a'}|$ between the boundary's two aldermen. The estimates are therefore per standard deviation of score gap, and boundaries whose aldermen have similar scores receive little weight. Multifamily density is 4.4% lower per standard deviation (-0.045 , $SE = 0.022$); the estimate for buildings with five or more units is -0.051 ($SE = 0.053$), and the estimate for all construction remains close to zero (-0.002 , $SE = 0.030$).

Panel A: 1,000 ft inside the less-stringent side



Panel B: 1,000 ft inside the more-stringent side

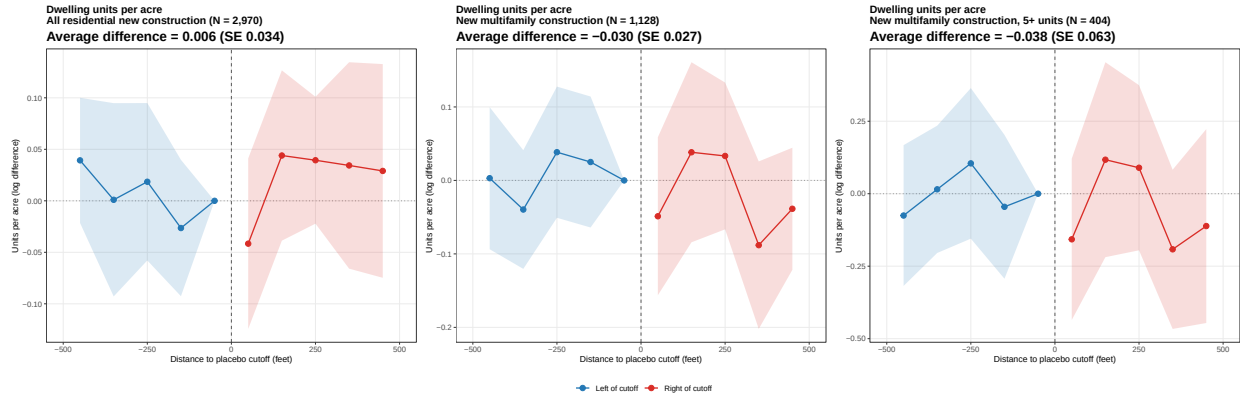
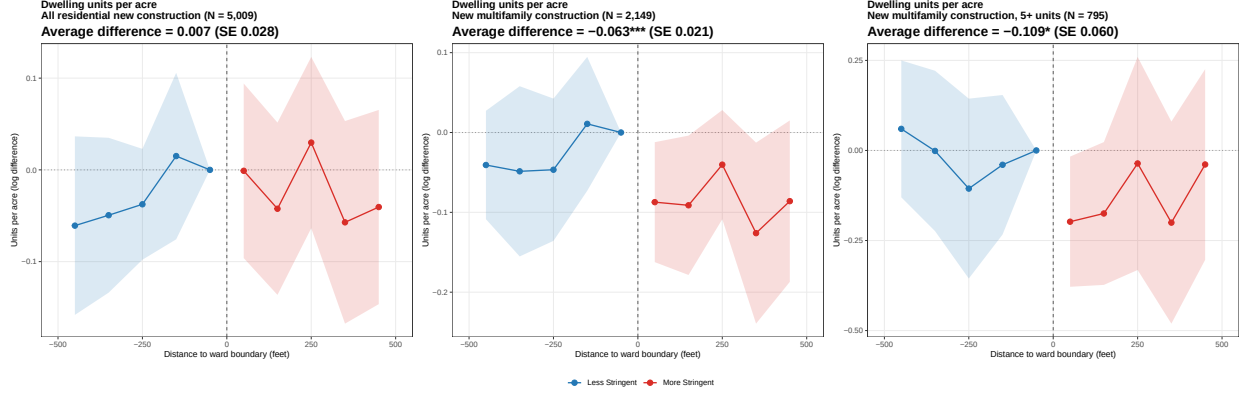


Figure D.1: Density Estimates at Placebo Cutoffs

Notes: Each artificial cutoff lies 1,000 ft inside the indicated side of the true ward boundary. The model uses a 500 ft window around the artificial cutoff and the same 100 ft distance bands, controls, and fixed effects as Equation (4). The 100 ft band just below the artificial cutoff is the reference group. Subtitles report the difference for the 100 ft band just above the cutoff and the average difference between the two sides of the window. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Excluding buildings within 25 ft



Panel B: Excluding buildings within 50 ft

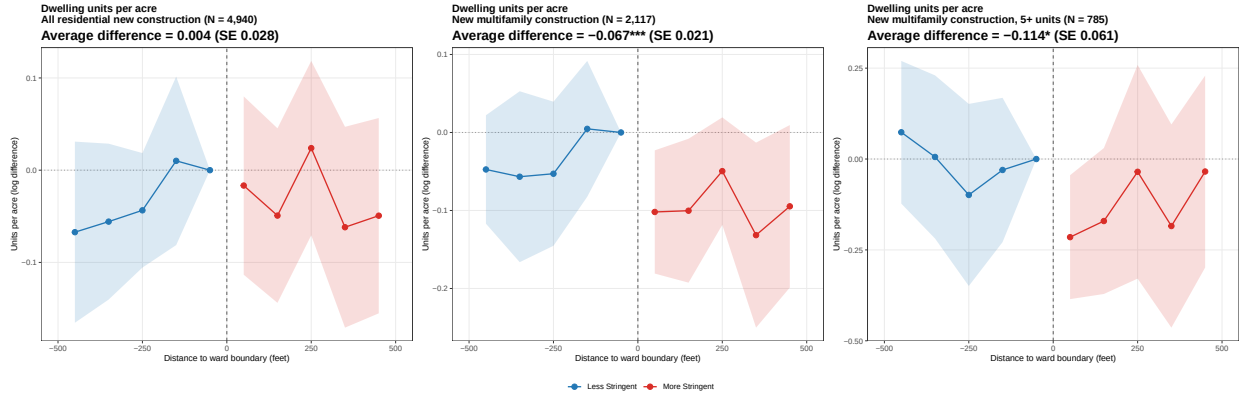


Figure D.2: Density Estimates Excluding Buildings Nearest the Boundary

Notes: This check asks whether buildings immediately adjacent to a boundary drive the main result. Each panel excludes buildings within the indicated distance and otherwise uses the 500 ft sample and Equation (4). After each exclusion, the remaining part of the -100 to 0 ft band on the less-stringent side is the reference group. Subtitles report the difference across the boundary and the average difference between the two sides of the window. In each panel, the multifamily estimates remain negative and slightly larger than in the full sample, and the all-construction estimates remain close to zero. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

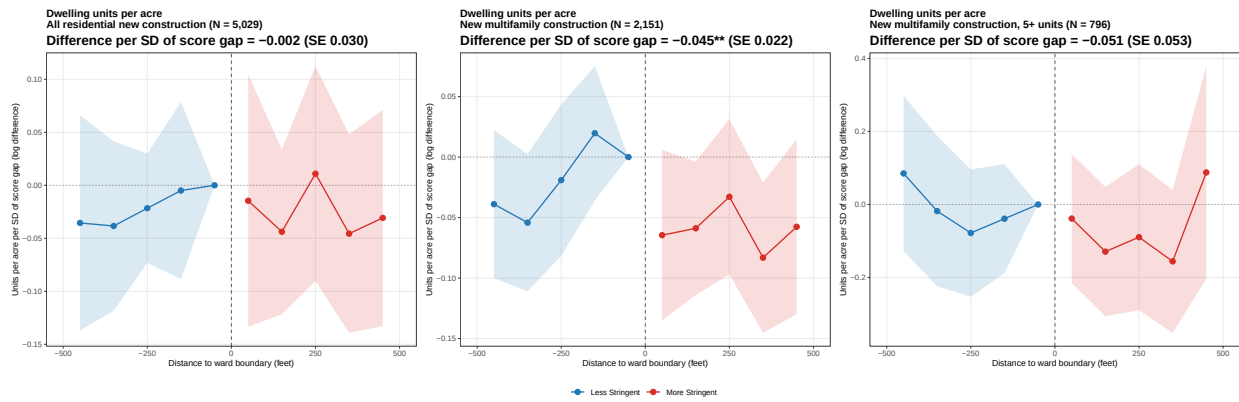


Figure D.3: Density at Ward Boundaries, Continuous Stringency Difference

Notes: The figure repeats Figure 3 with each distance band multiplied by the absolute score gap between the two aldermen. Points show differences in log dwelling units per acre per standard deviation of score gap, relative to the band from -100 to 0 ft. Subtitles report the average difference per standard deviation of score gap. Shaded areas are 95% confidence intervals based on standard errors clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.1: Density Estimates Under Alternative Measures of Boundary Stringency

	All Construction	Multifamily	Multifamily, 5+ Units
Panel A: Removing permits linked to each building			
<i>Full-sample score</i>			
Difference across boundary	-0.021 (0.051)	-0.084** (0.038)	-0.184** (0.091)
Average difference	0.002 (0.029)	-0.061*** (0.021)	-0.100* (0.059)
Observations	5,029	2,151	796
Ward pairs/segments	124/928	96/600	71/387
<i>Score excluding the building's permits</i>			
Difference across boundary	-0.021 (0.051)	-0.084** (0.038)	-0.184** (0.091)
Average difference	0.002 (0.029)	-0.061*** (0.021)	-0.100* (0.059)
Observations	5,029	2,151	796
Ward pairs/segments	124/928	96/600	71/387
Panel B: Excluding boundaries with similar scores			
<i>Score difference at least 0.25 SD</i>			
Difference across boundary	-0.031 (0.066)	-0.072 (0.044)	-0.105 (0.130)
Average difference	-0.002 (0.043)	-0.043 (0.026)	-0.034 (0.064)
Observations	3,751	1,674	608
Ward pairs/segments	120/824	90/506	66/309
<i>Score difference at least 0.50 SD</i>			
Difference across boundary	-0.053 (0.063)	-0.047 (0.051)	-0.033 (0.161)
Average difference	-0.032 (0.040)	-0.026 (0.028)	-0.049 (0.068)
Observations	2,966	1,295	487
Ward pairs/segments	105/652	78/410	59/250

Notes: The outcome is log dwelling units per acre, estimated with Equation (4). The difference across the boundary compares the two 100 ft bands immediately across it; the average difference compares the two sides of the 500 ft window. In Panel A, the second set of rows removes the permits matched to each building before recalculating the alderman scores used for that building. Panel B retains boundaries where the absolute difference between the two alderman scores is at least the stated number of standard deviations. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.2: Continuity in Location and Neighborhood Characteristics at Ward Boundaries

	All Construction	Multifamily	Multifamily, 5+ Units
<i>Location</i>			
Distance to downtown (miles)	-0.002 (0.007)	-0.011 (0.007)	-0.002 (0.012)
Distance to nearest school (miles)	0.006 (0.004)	0.000 (0.005)	0.001 (0.007)
Distance to nearest park (miles)	-0.006 (0.006)	-0.006 (0.006)	-0.006 (0.009)
Distance to Lake Michigan (miles)	-0.003 (0.006)	-0.010** (0.004)	0.010 (0.009)
<i>2000 census block</i>			
Non-Hispanic white share	-0.053*** (0.019)	-0.041 (0.030)	-0.112 (0.071)
Non-Hispanic Black share	-0.012** (0.006)	-0.006 (0.009)	-0.009 (0.028)
Owner-occupied share	0.028 (0.022)	0.036 (0.032)	0.099 (0.085)
Vacancy rate	0.007 (0.005)	0.001 (0.010)	0.004 (0.017)
Housing units per acre	-0.126 (1.046)	-0.021 (0.879)	0.131 (1.526)
Observations	5,029	2,151	796

Notes: Each row uses the listed characteristic as the dependent variable. The census-block rows use the 2000 decennial census block containing each building; the race shares, owner-occupied share, and vacancy rate omit buildings in blocks without residents, occupied units, or housing units. The reported coefficient compares the 0–100 ft band on the more-stringent side with the –100 to 0 ft band on the less-stringent side. Regressions include 100 ft distance-band indicators and boundary-segment-by-joint-service fixed effects. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.3: Density Estimates Along Different Types of Ward Boundaries and With Segment-by-Year Fixed Effects

	All Construction	Multifamily	Multifamily, 5+ Units
<i>Main sample</i>			
Difference across boundary	-0.021 (0.051)	-0.084** (0.038)	-0.184** (0.091)
Average difference	0.002 (0.029)	-0.061*** (0.021)	-0.100* (0.059)
Observations	5,029	2,151	796
<i>Limited expressway or water overlap</i>			
Difference across boundary	-0.023 (0.052)	-0.085** (0.038)	-0.185** (0.092)
Average difference	-0.002 (0.030)	-0.062*** (0.021)	-0.104* (0.061)
Observations	4,753	2,095	766
<i>Limited physical-feature or arterial overlap</i>			
Difference across boundary	-0.013 (0.054)	-0.065* (0.036)	-0.228** (0.090)
Average difference	0.005 (0.033)	-0.037* (0.021)	-0.120* (0.066)
Observations	4,033	1,743	615
<i>Straight boundary segment</i>			
Difference across boundary	0.005 (0.052)	-0.093** (0.041)	-0.207* (0.113)
Average difference	0.033 (0.029)	-0.053* (0.027)	-0.086 (0.080)
Observations	3,871	1,551	576
<i>Segment $\times$ construction-year fixed effects</i>			
Difference across boundary	-0.033 (0.063)	-0.133** (0.063)	-0.141 (0.120)
Average difference	-0.016 (0.033)	-0.105*** (0.028)	-0.069 (0.065)
Observations	5,029	2,151	796

Notes: The outcome is log dwelling units per acre, estimated with Equation (4). The difference across the boundary compares the two 100 ft bands immediately across it; the average difference compares the two sides of the 500 ft window. The expressway-or-water restriction removes segments for which either feature overlaps at least half of the segment. The broader restriction removes segments with at least 50% park, water, or cemetery overlap, 40% expressway overlap, or 75% arterial-road overlap. A boundary is classified as locally straight when a 328 ft straight line centered on the building's nearest boundary point stays within 49 ft of the actual boundary at both ends. The last block uses the full sample with segment-by-construction-year and zoning-group fixed effects in place of segment-by-joint-service fixed effects. Standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Additional Price Boundary Checks

This appendix reports the price comparisons at placebo cutoffs (Figure E.1), on locally straight boundaries and excluding observations nearest the boundary (Figure E.2), and with the continuous stringency difference (Figure E.3).

Prices at artificial boundaries

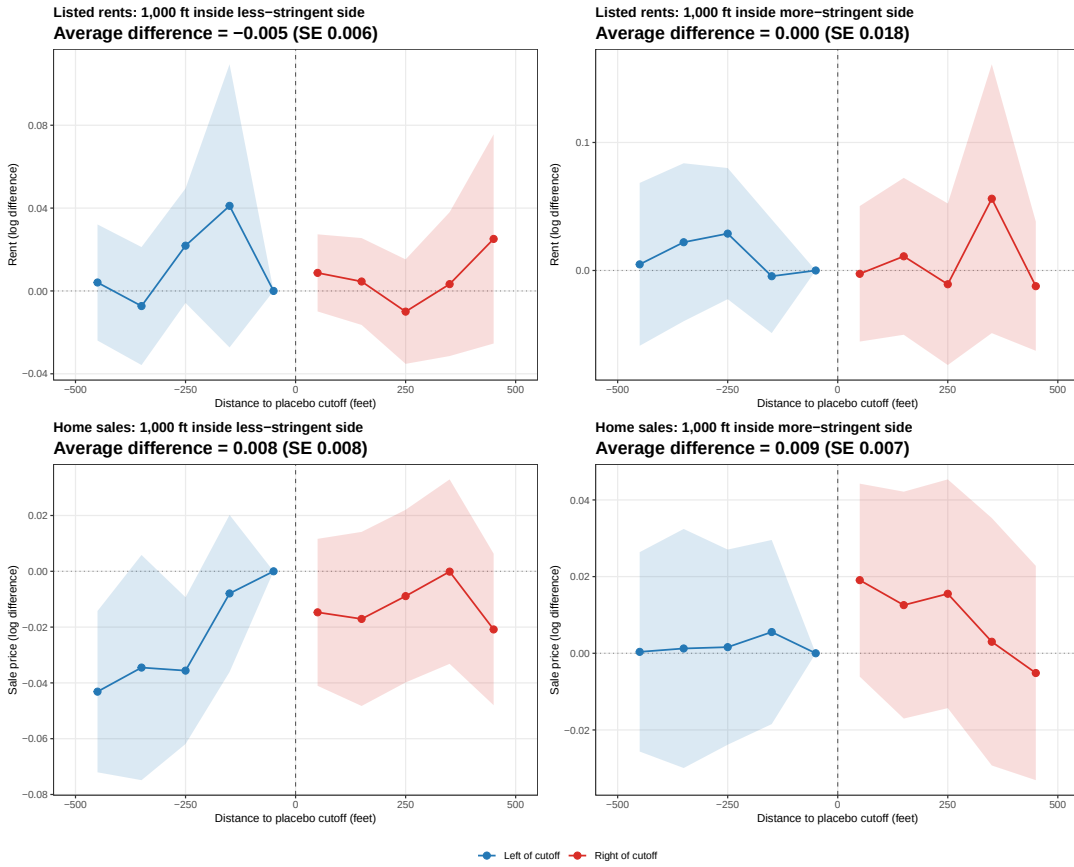
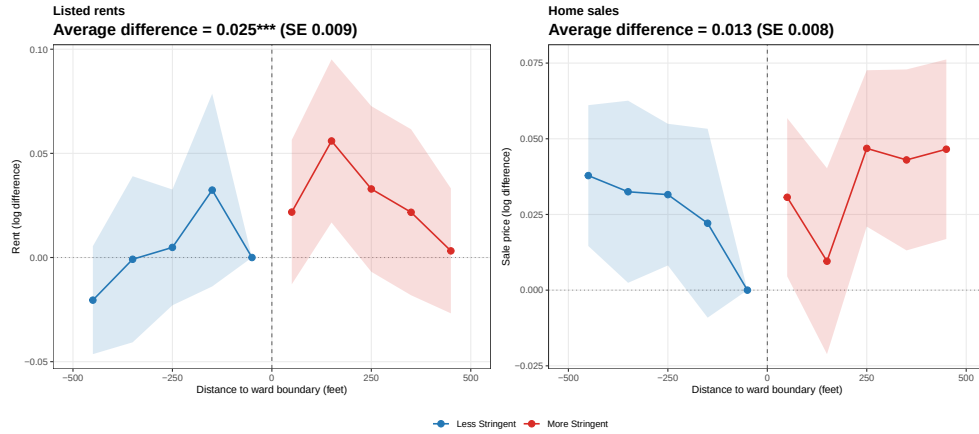


Figure E.1: Placebo Cutoffs for Listed Rents and Home Sale Prices

Notes: Neither rents nor sale prices change significantly across an artificial cutoff. The left column places an artificial cutoff 1,000 ft inside the less-stringent ward; the right column places it 1,000 ft inside the more-stringent ward. The top row reports listed rents and the bottom row reports home sale prices. Each model uses a 500 ft window around the artificial cutoff and otherwise follows Equation (5). Each point is the estimated log difference between a 100 ft distance band and the band immediately below the artificial cutoff. Subtitles report the difference between the bands immediately above and below the cutoff and the average difference, which replaces the distance bands with an indicator for the side of the window above the cutoff. Shaded areas are 95% confidence intervals. Standard errors are clustered by ward pair. *

$p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Locally straight boundaries



Panel B: Excluding observations nearest the boundary

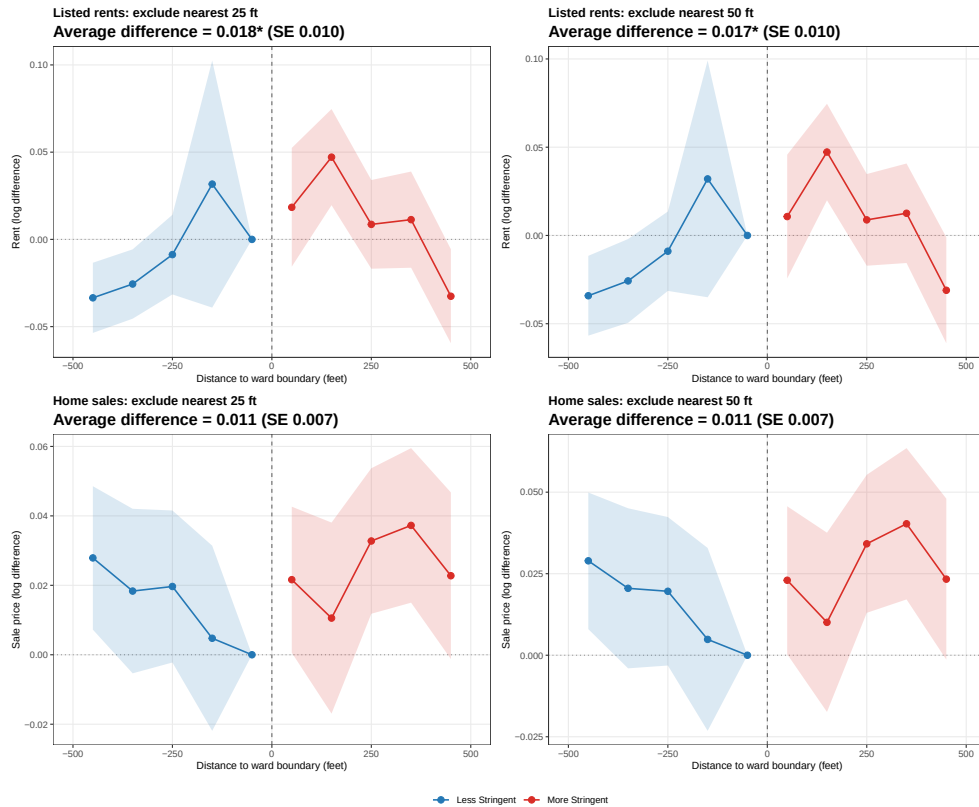


Figure E.2: Additional Price-Boundary Robustness Checks

Notes: Panel A retains locally straight boundaries: a 328 ft line centered on the nearest boundary point stays within 49 ft of the boundary at both ends. Panel B excludes observations within 25 ft (left) or 50 ft (right); the top row reports listed rents and the bottom row home sale prices. All models otherwise follow Equation (5). Points report log differences from the nearest 100 ft band on the less-stringent side, normalized to zero. Subtitles report the difference between the bands immediately across the boundary and the average difference, which replaces the distance bands with an indicator for the more-stringent side of the 500 ft window. Panel B uses the remaining observations in each band after exclusion. Shaded areas are 95% confidence intervals, and standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

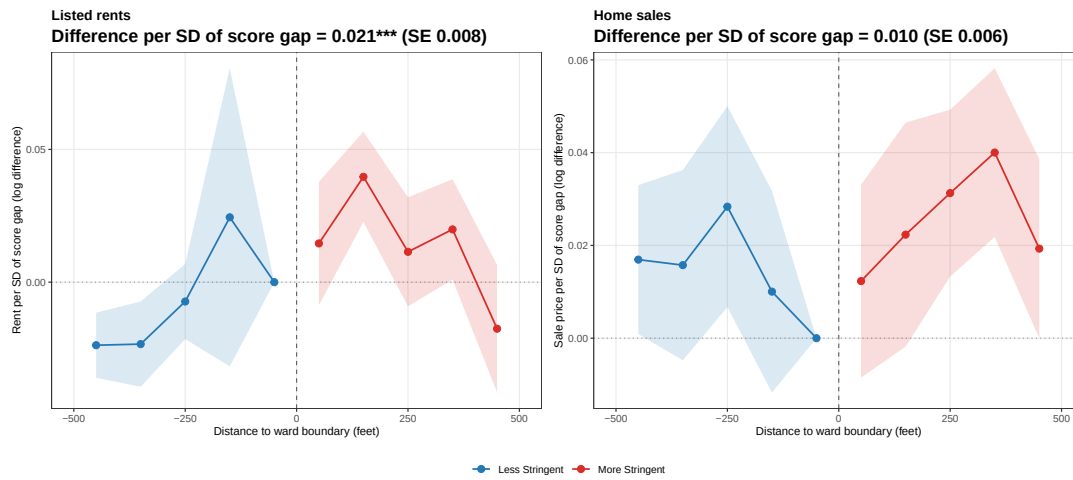


Figure E.3: Listed Rents and Home Sale Prices at Ward Boundaries, Continuous Stringency Difference

Notes: The figure repeats Figure 4 with each distance band, and the more-stringent-side indicator behind the subtitle, multiplied by the absolute score gap between the two aldermen. Points show log differences per standard deviation of score gap, relative to the nearest 100 ft band on the less-stringent side. Subtitles report the average difference per standard deviation of score gap: 0.021 for rents and 0.010 for sale prices. All models otherwise follow Equation (5). Shaded areas are 95% confidence intervals, and standard errors are clustered by ward pair. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.